

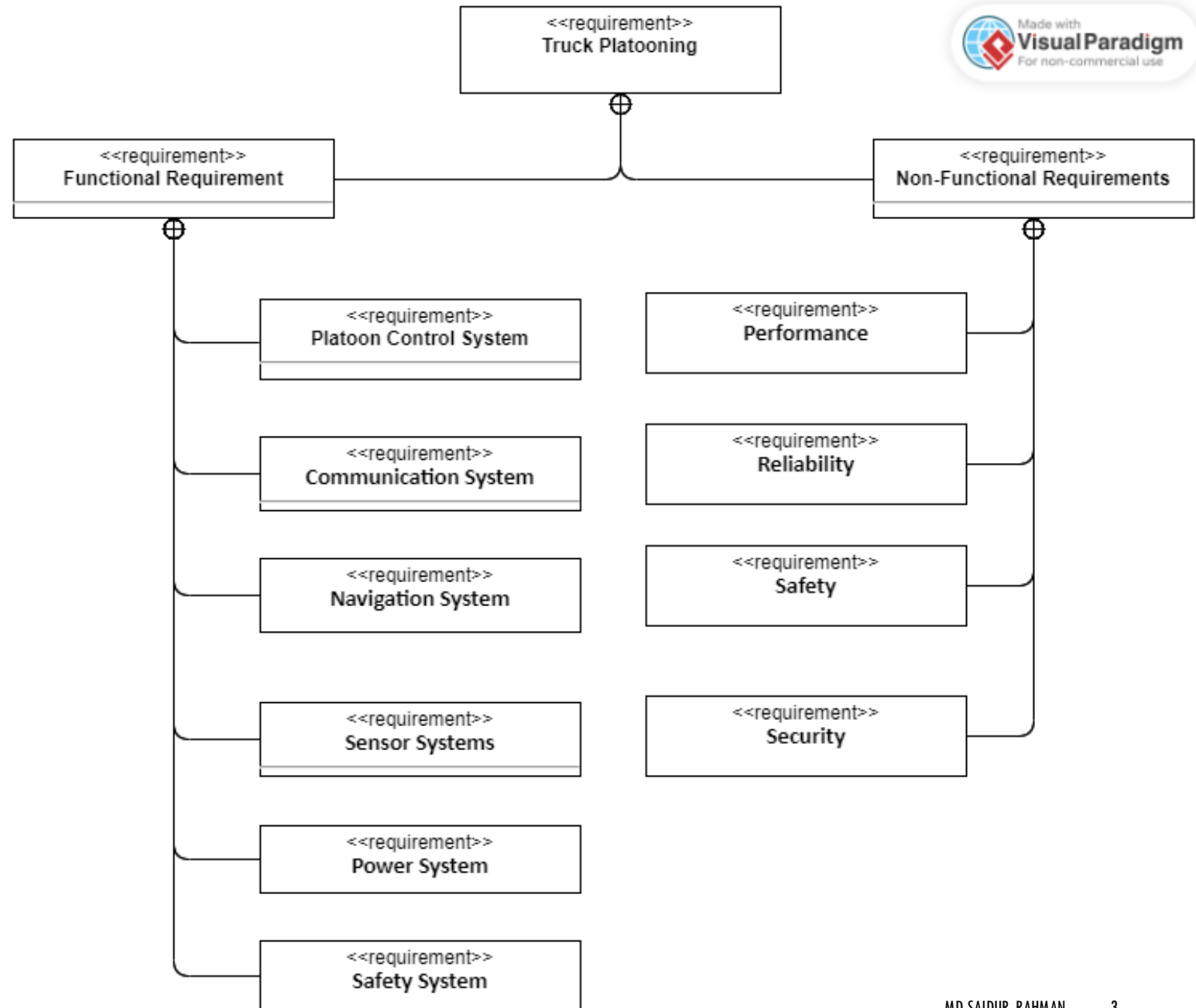


TRUCK PLATOONING

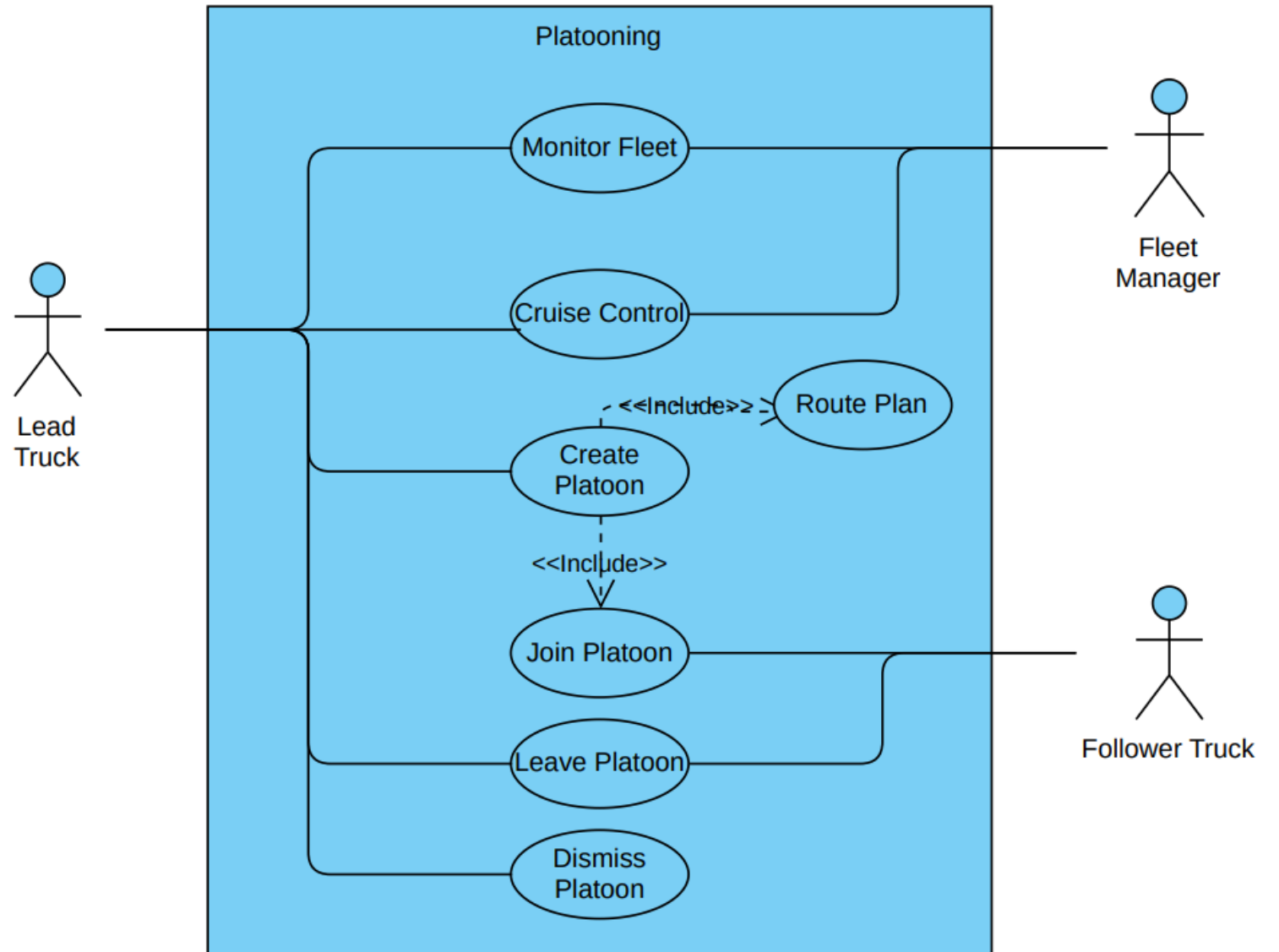
Truck Net (Group 3):

- Arsany Girgis
- Md Saidur Rahman
- Moshour Rahman Prince
- Sahat Al Ferdous Fahim

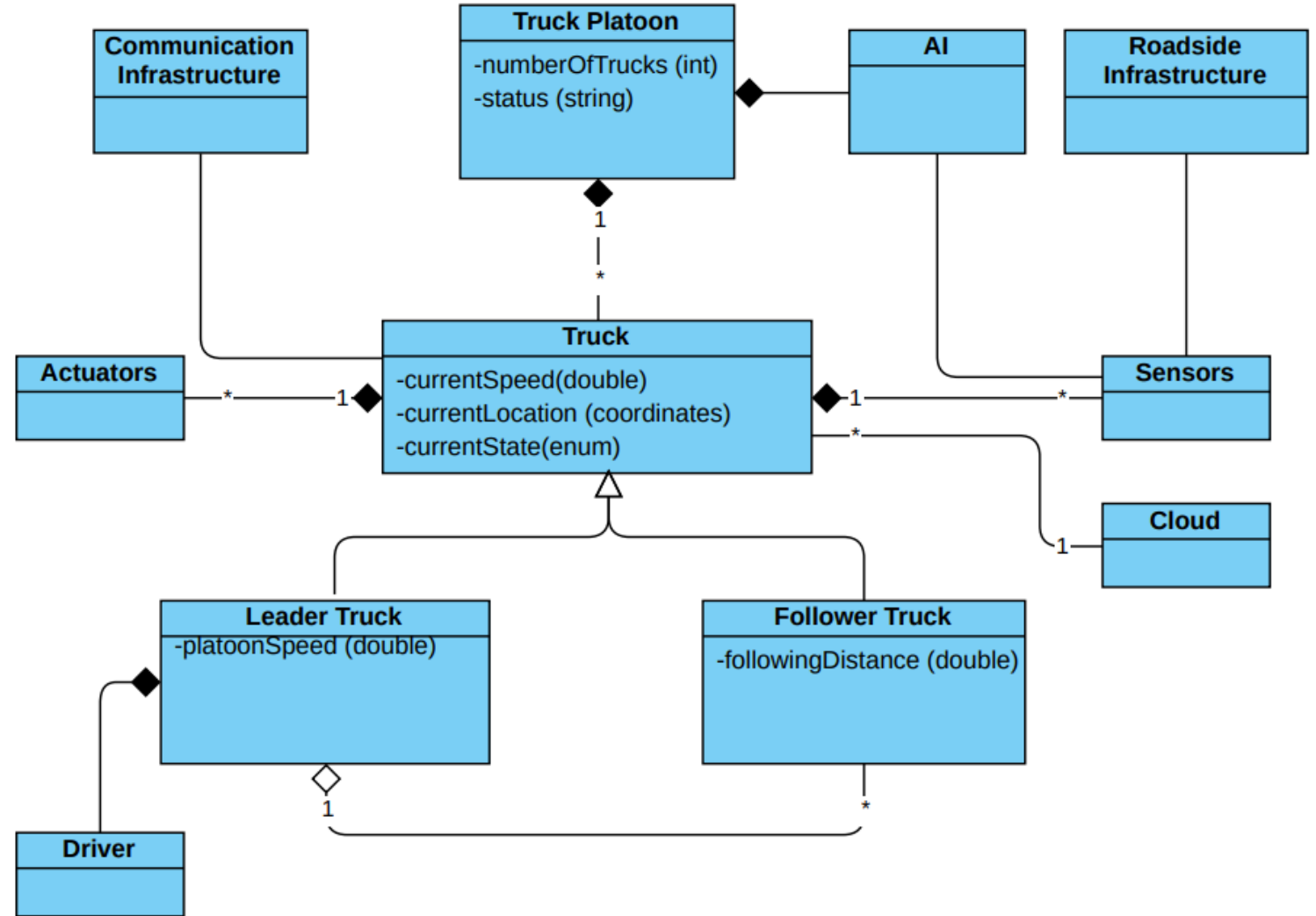
CONCEPT: *REQUIREMENTS*



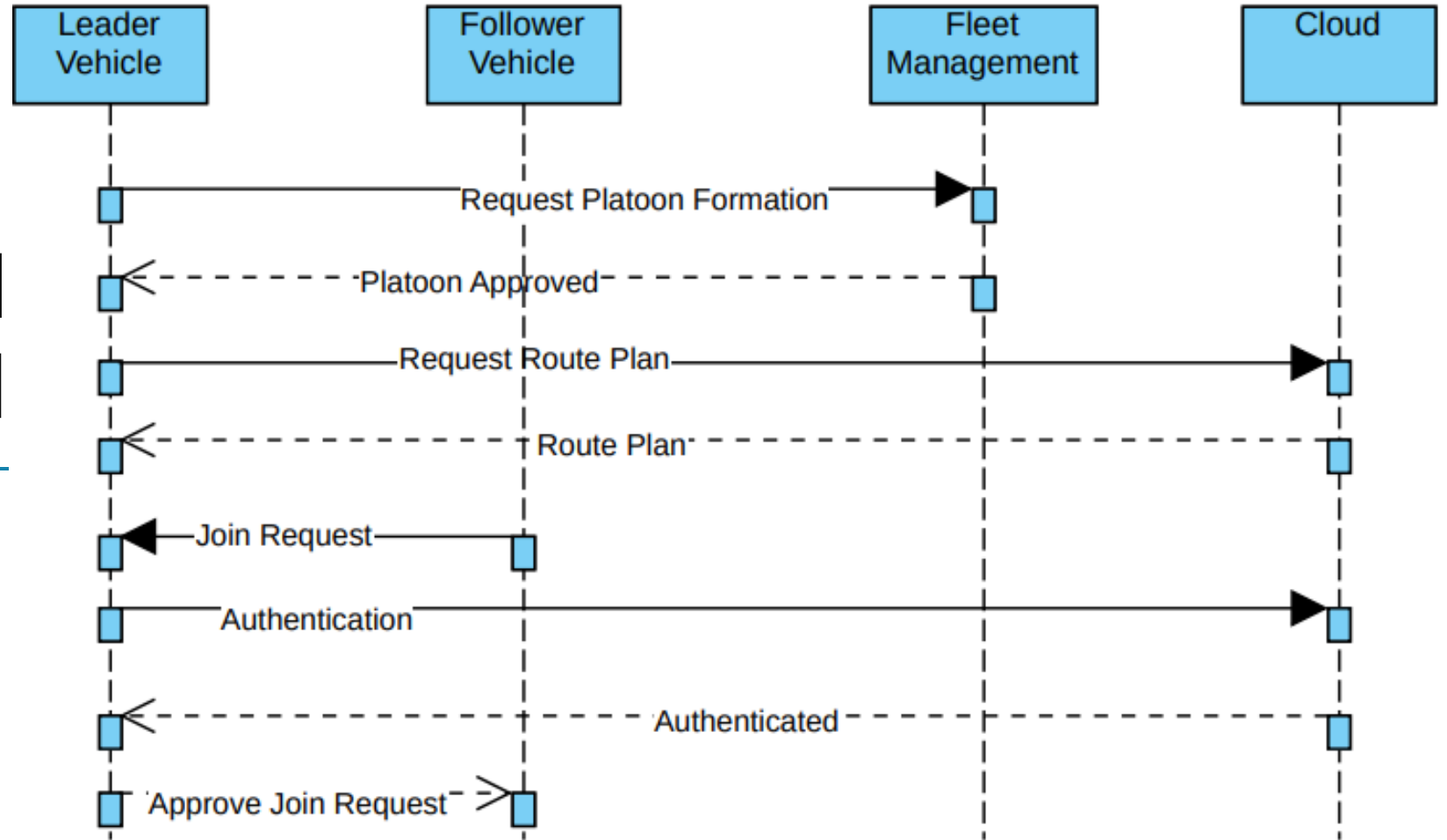
CONCEPT: *USE CASE*



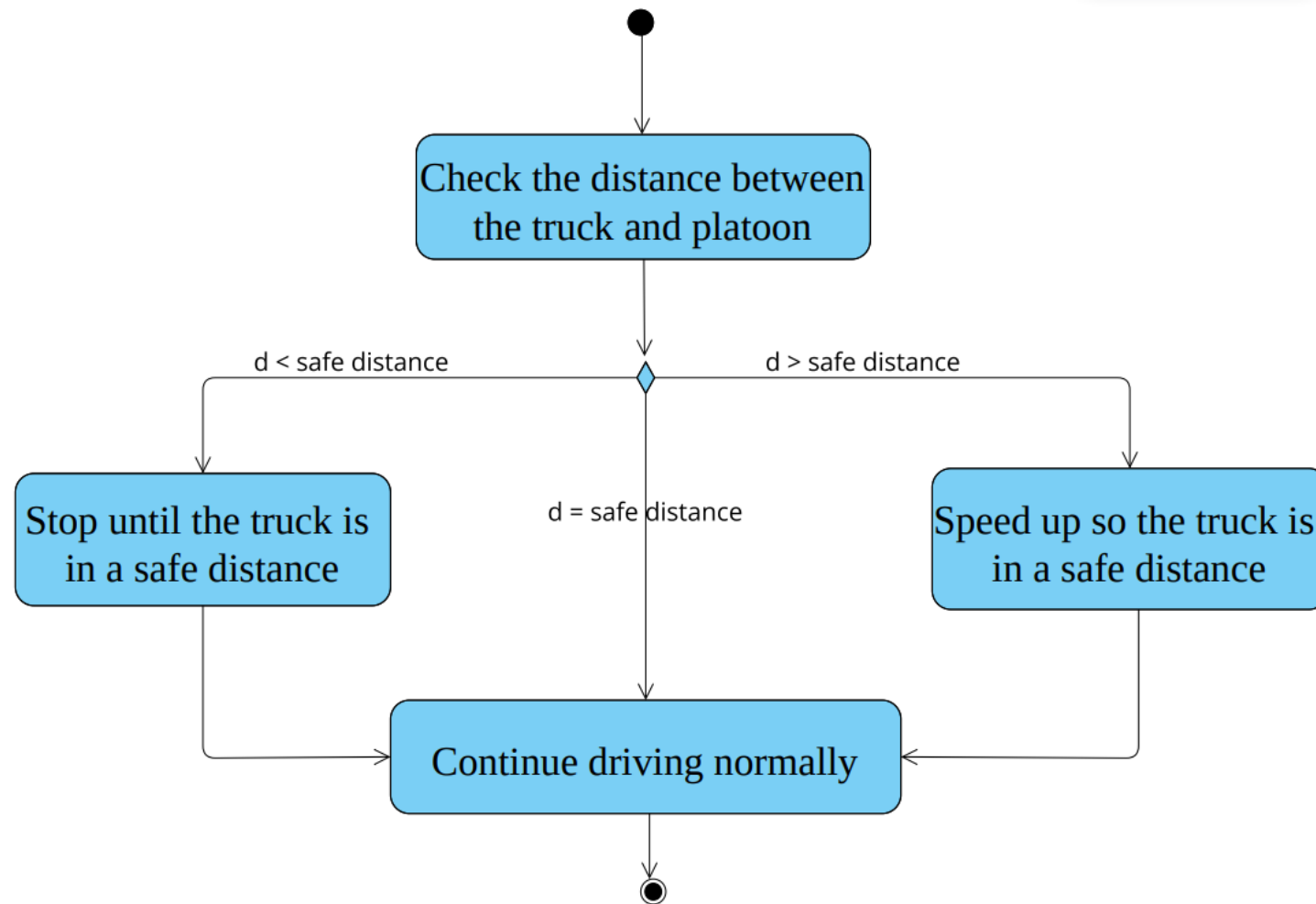
CLASS DIAGRAM



PLATOON FORMATION



SAFETY



COMMUNICATION

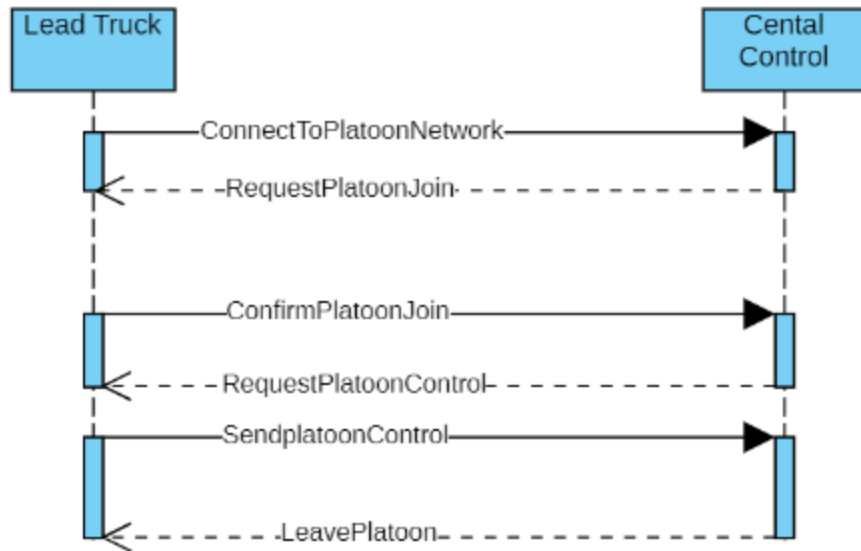


Fig: V2C communication

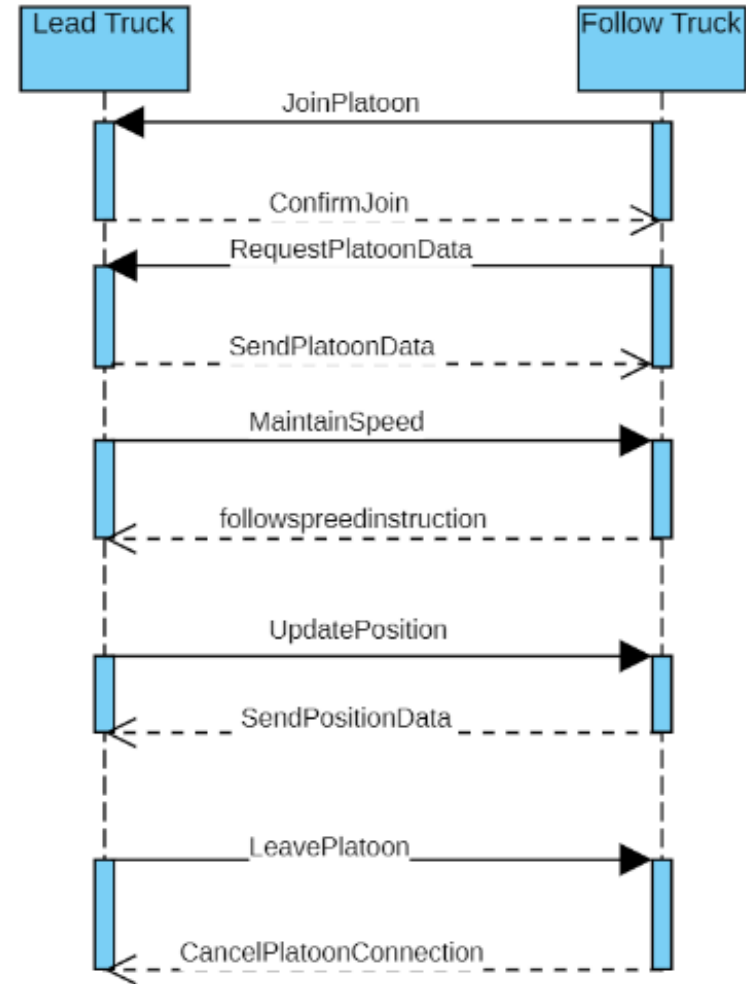
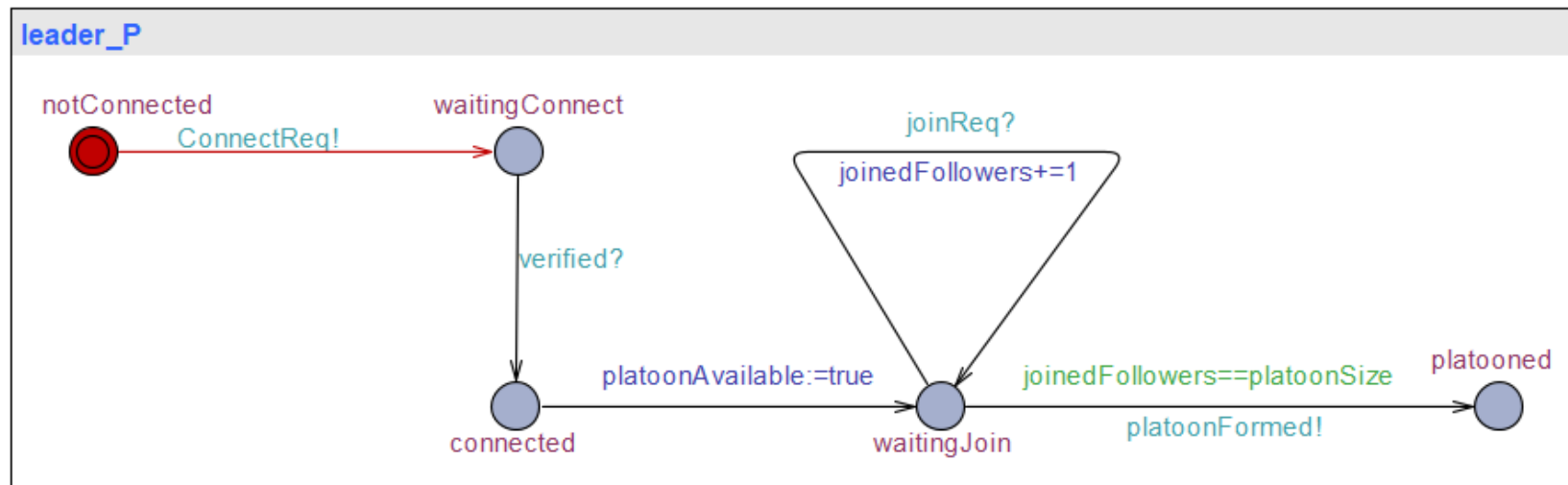
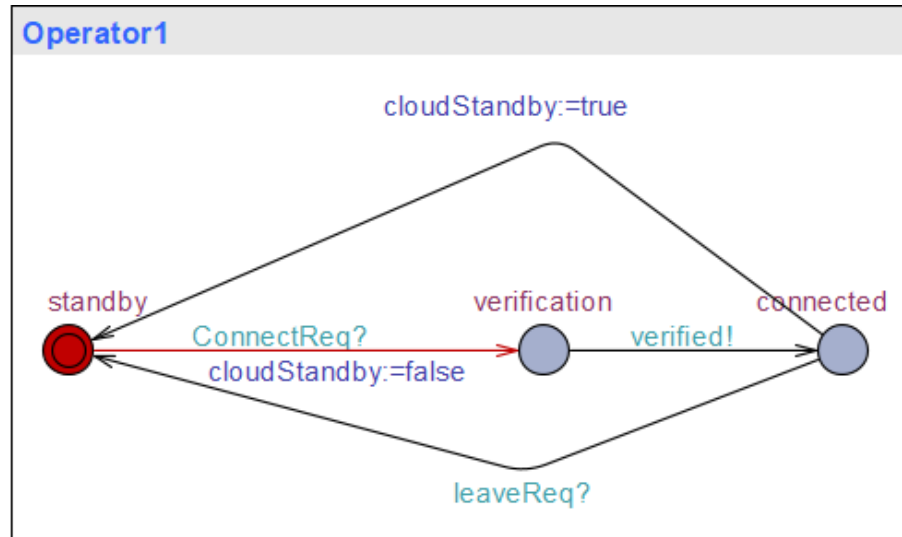
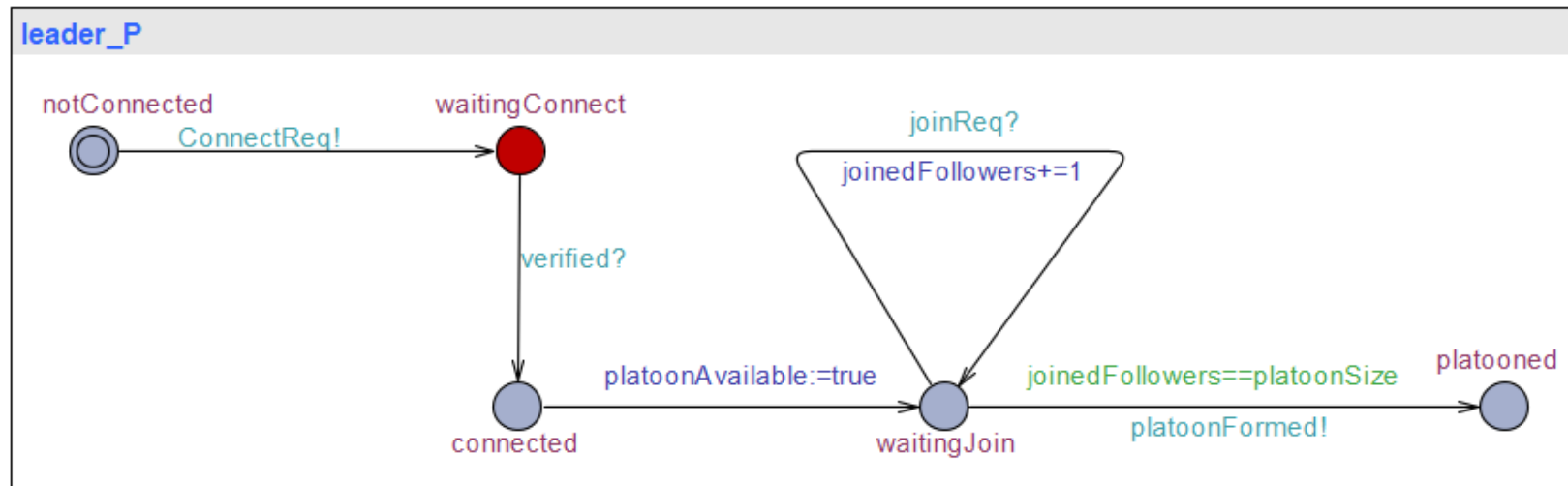
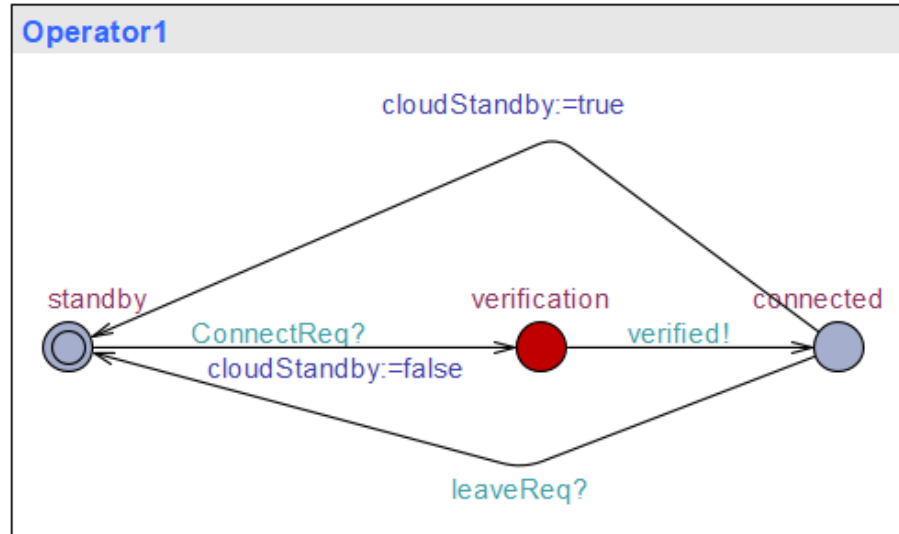


Fig: V2V communication

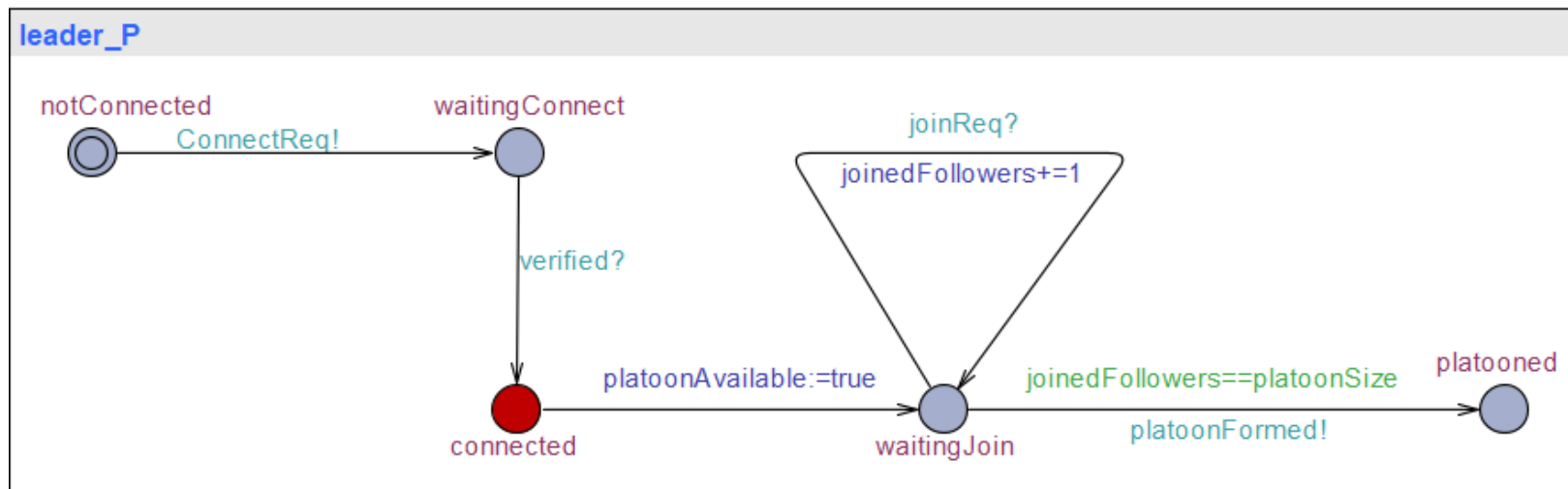
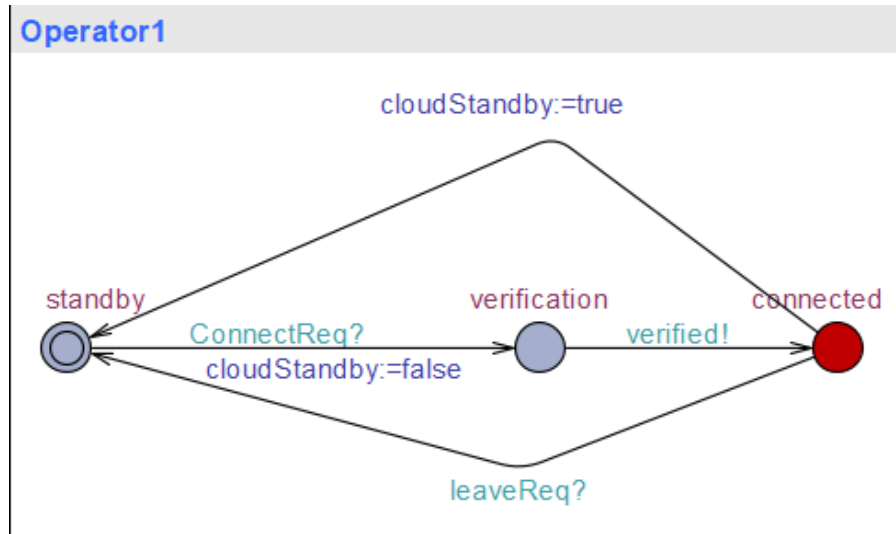
UPPAAL: FORMATION



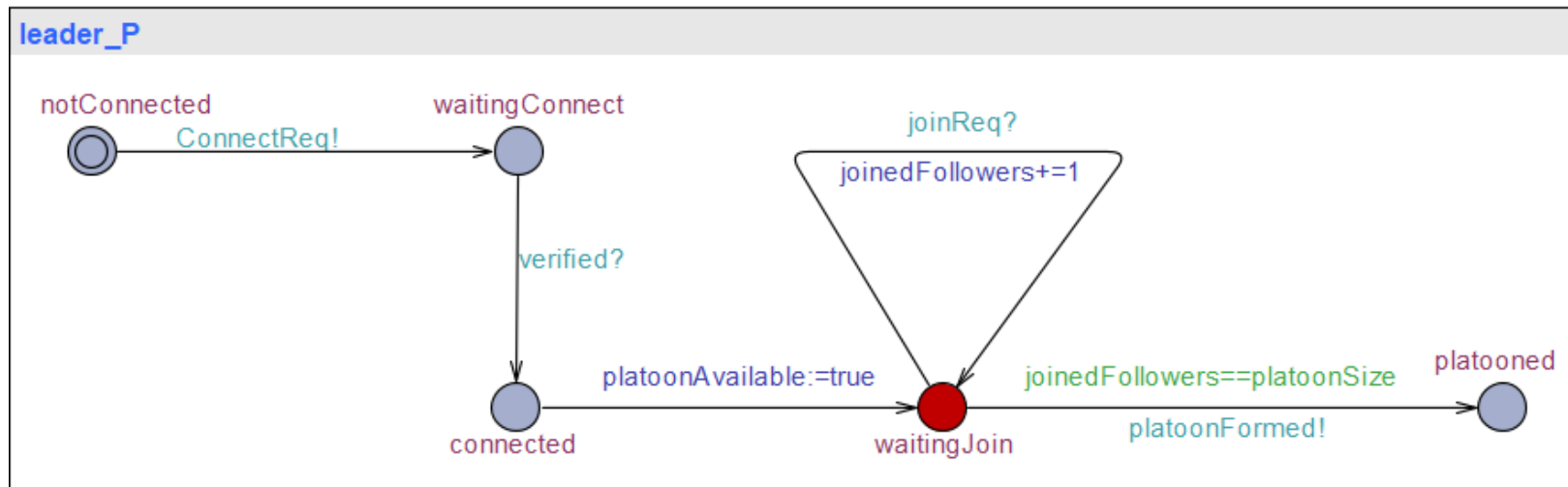
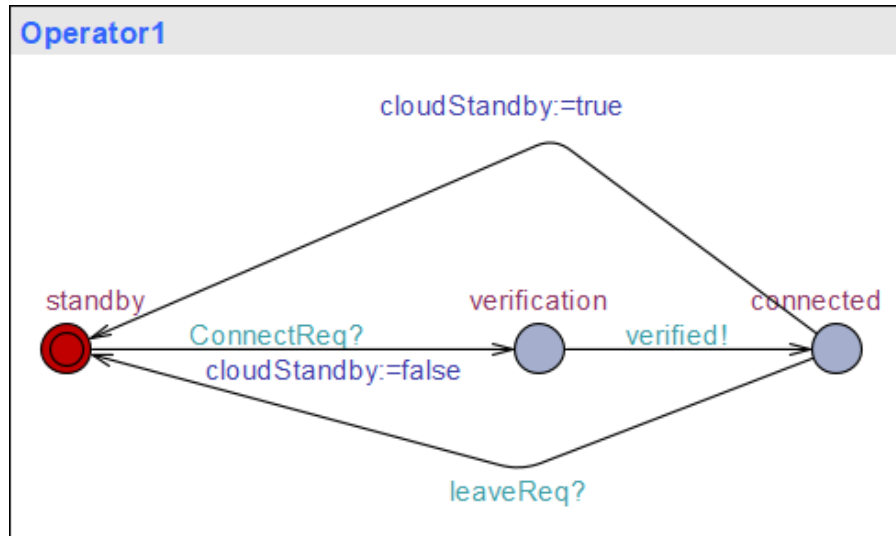
UPPAAL: FORMATION



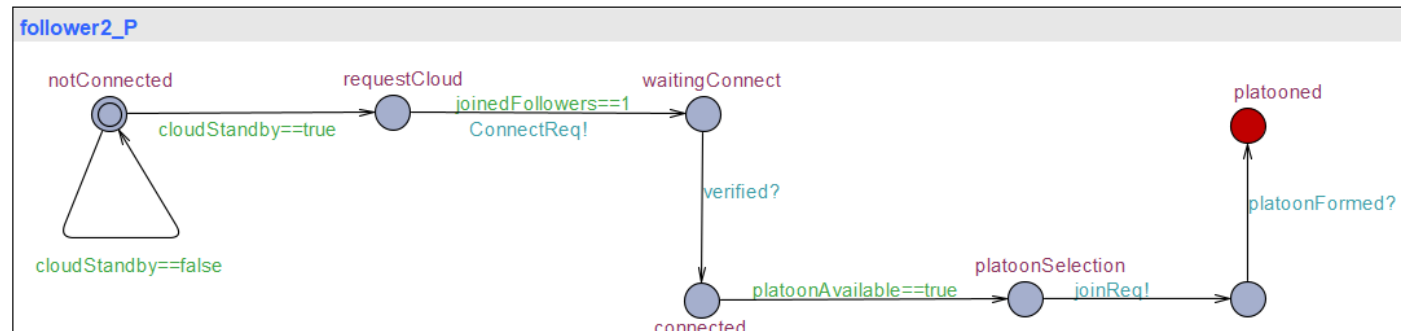
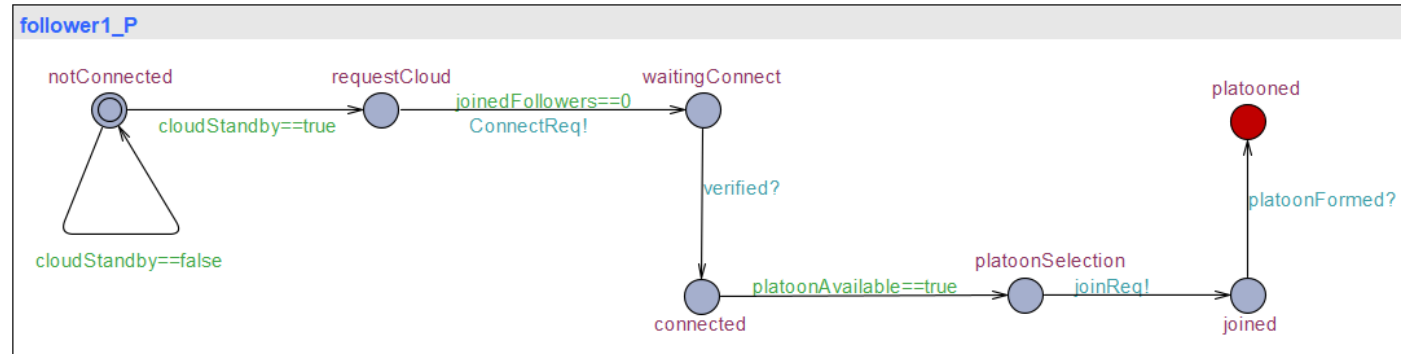
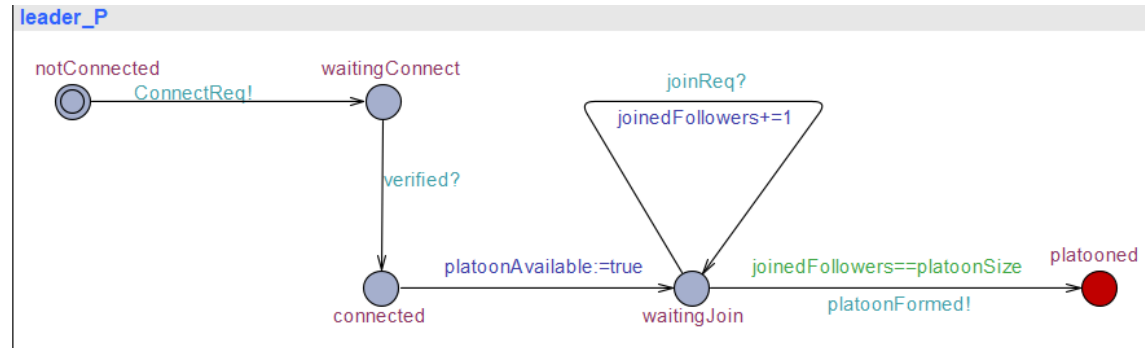
UPPAAL: FORMATION



UPPAAL: FORMATION

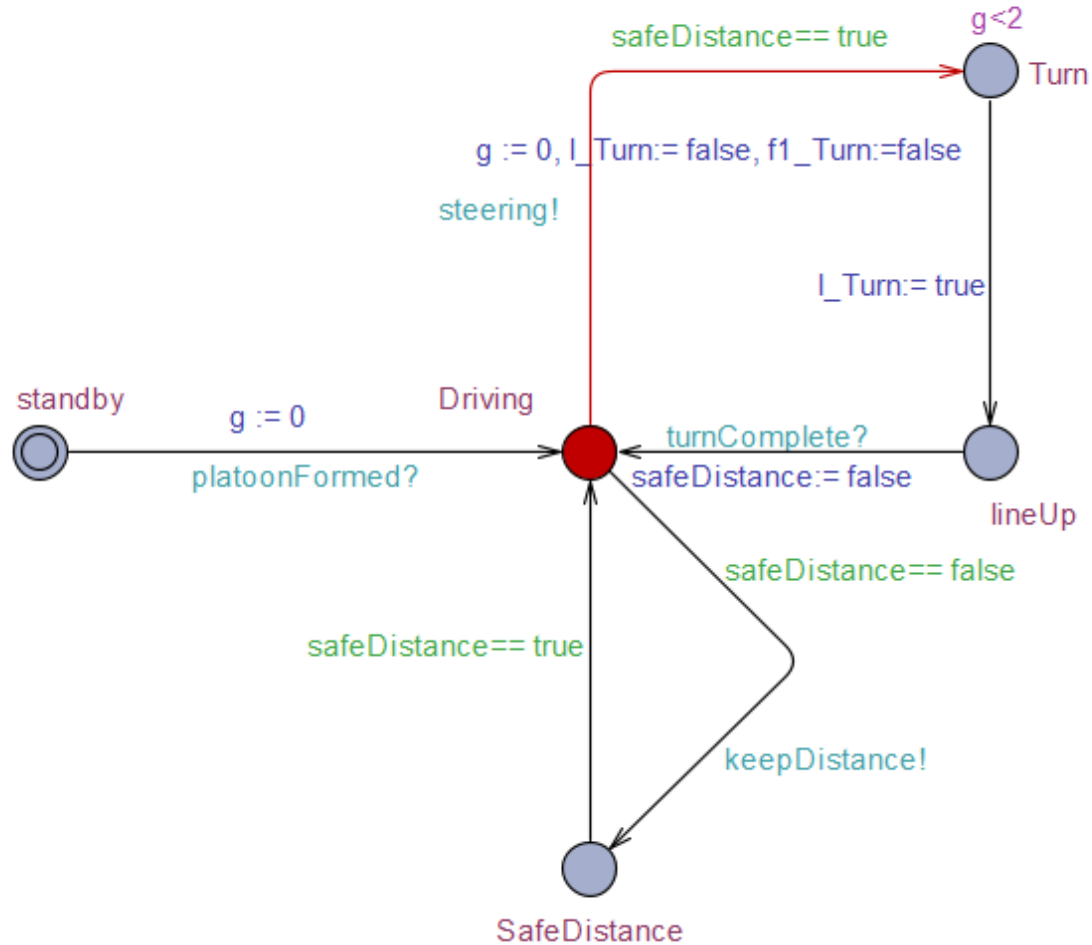


UPPAAL: FORMATION

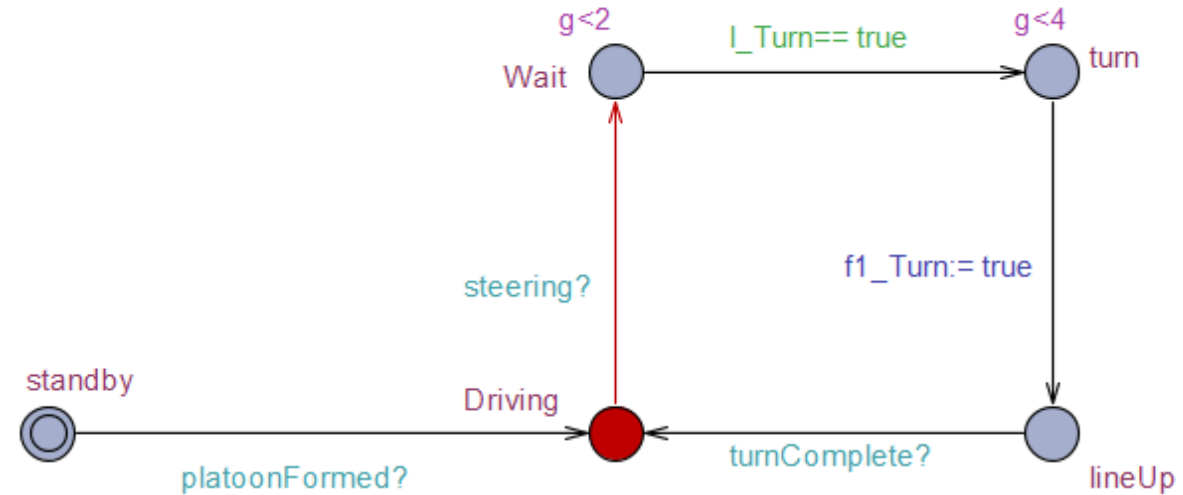


UPPAAL: FORMATION

leader_ACC

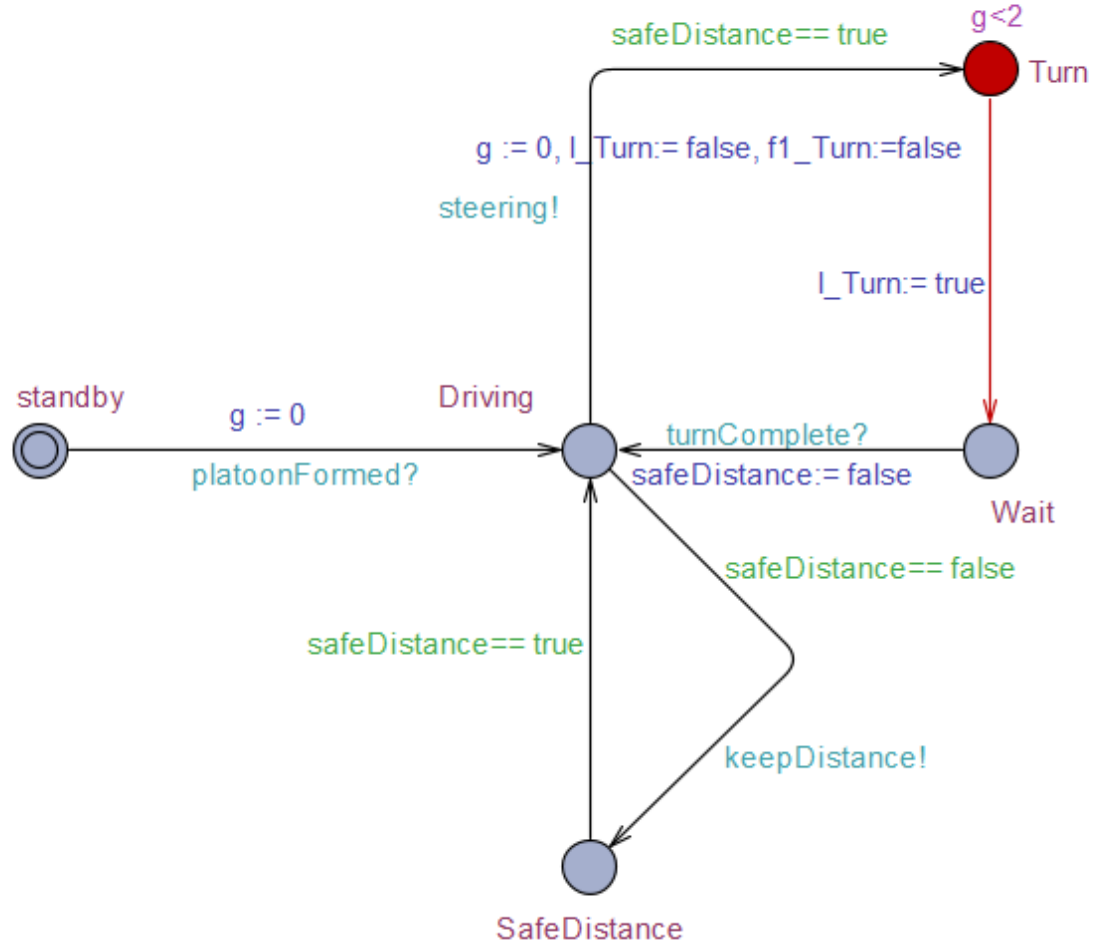


follower1_ACC

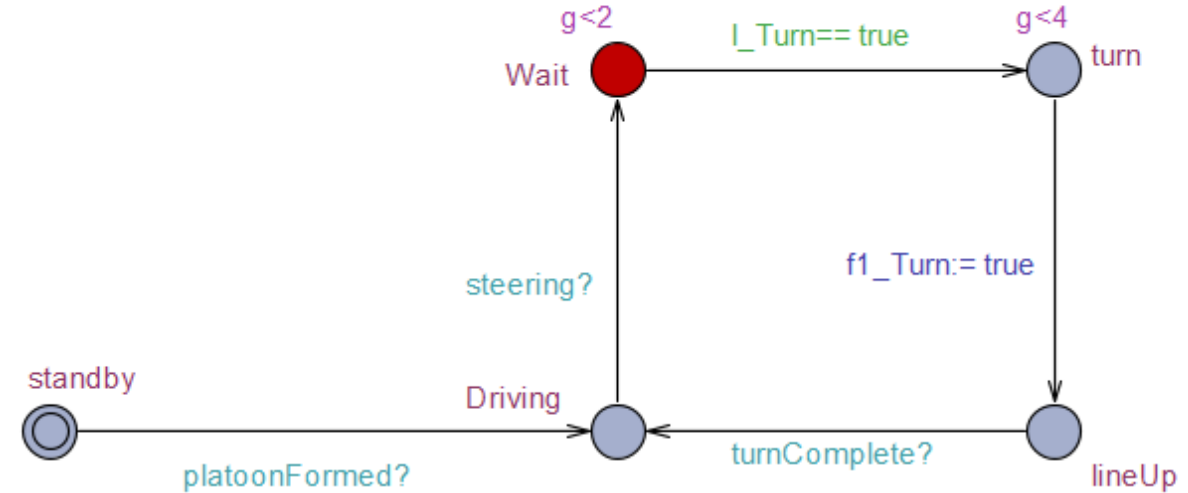


UPPAAL: FORMATION

leader_ACC

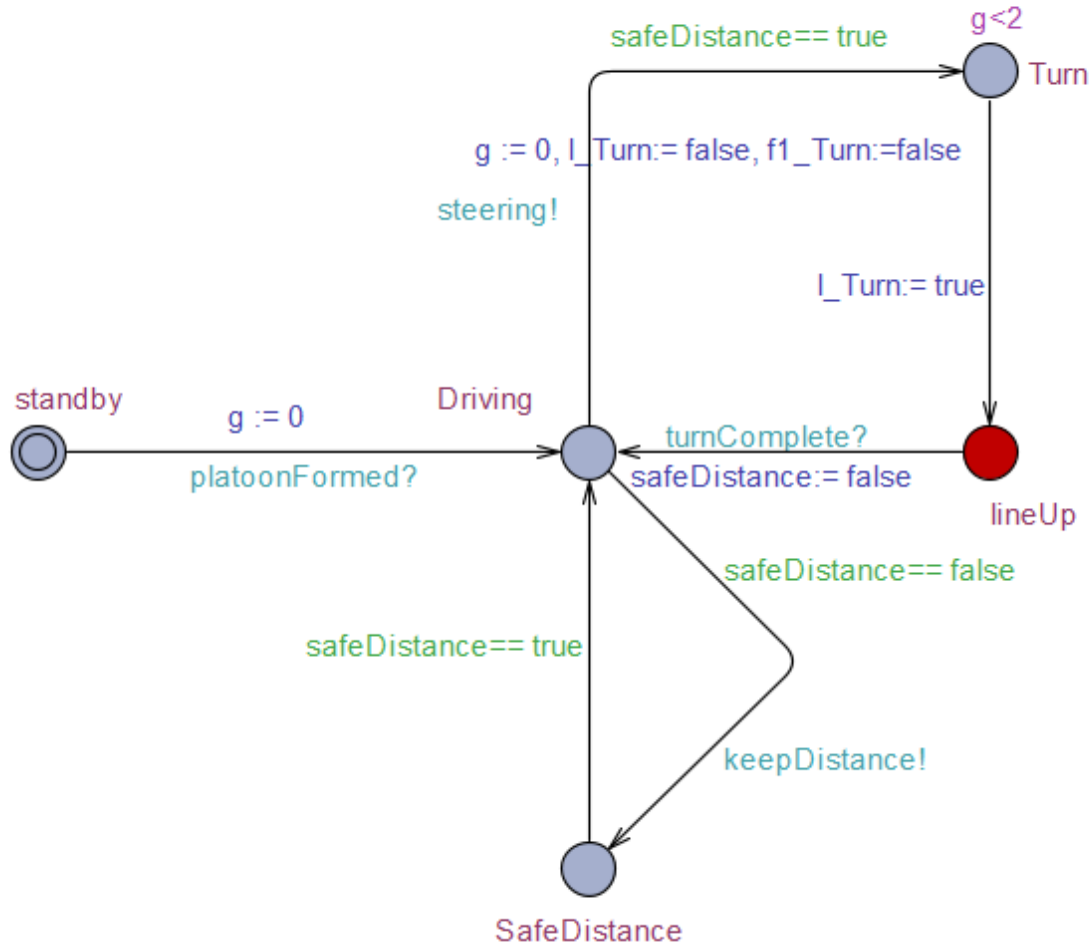


follower1_ACC

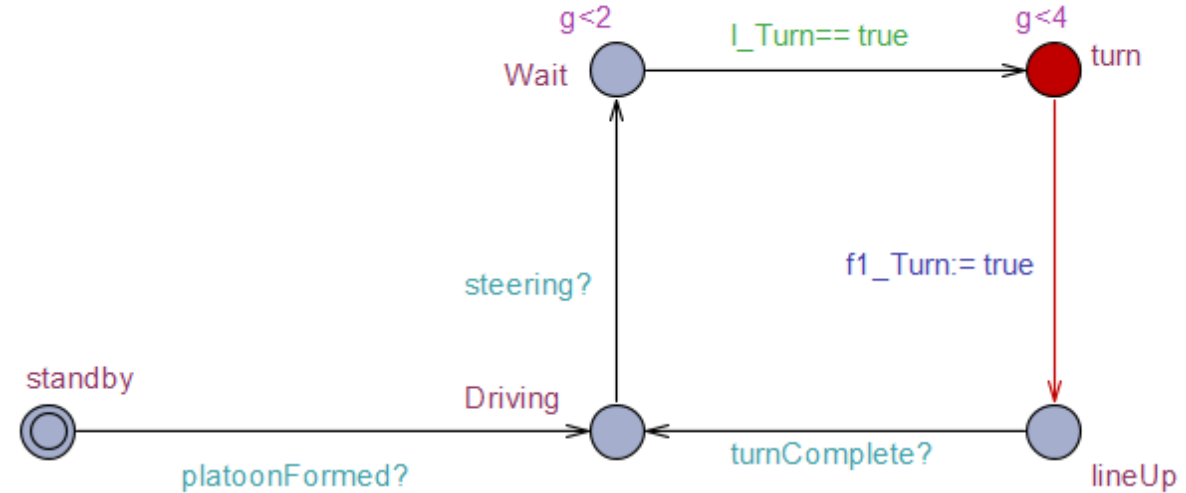


UPPAAL: FORMATION

leader_ACC

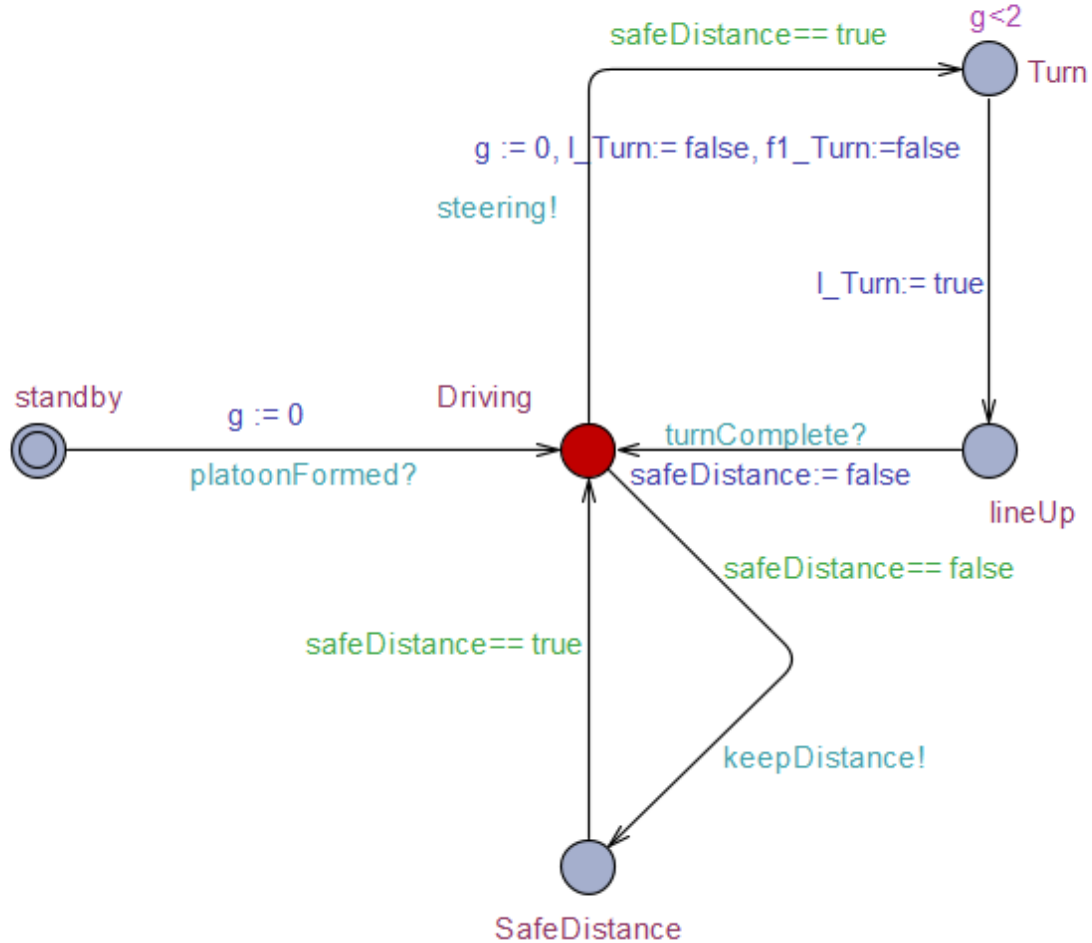


follower1_ACC

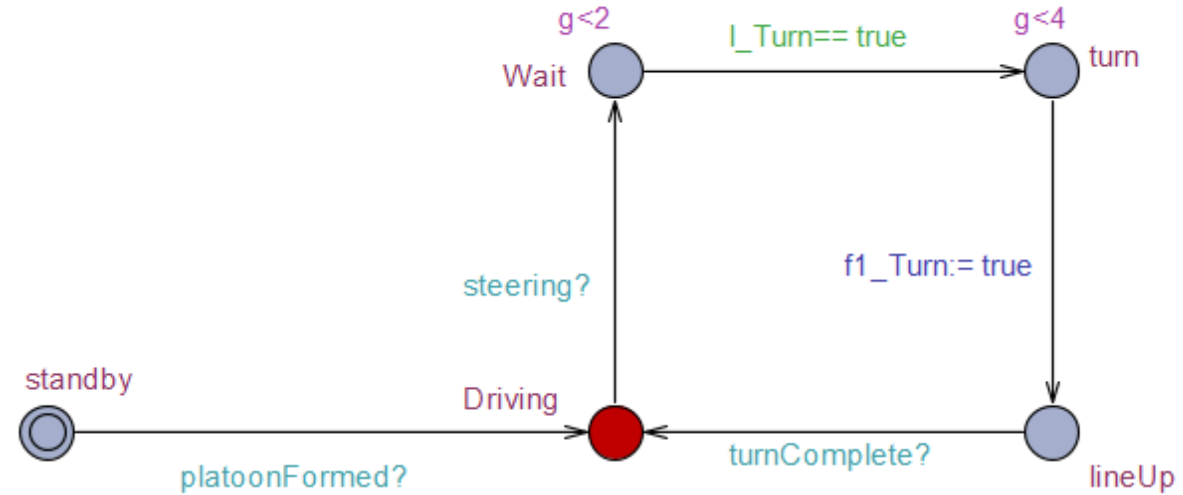


UPPAAL: FORMATION

leader_ACC

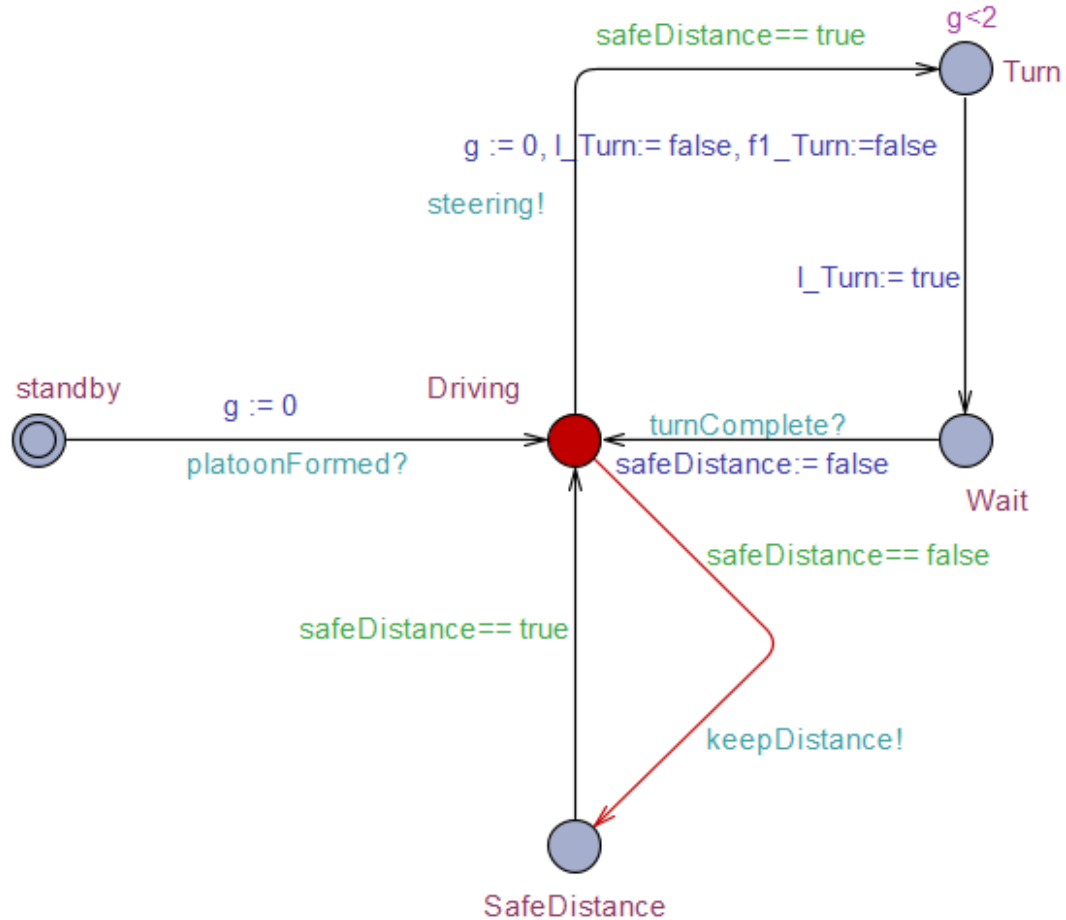


follower1_ACC

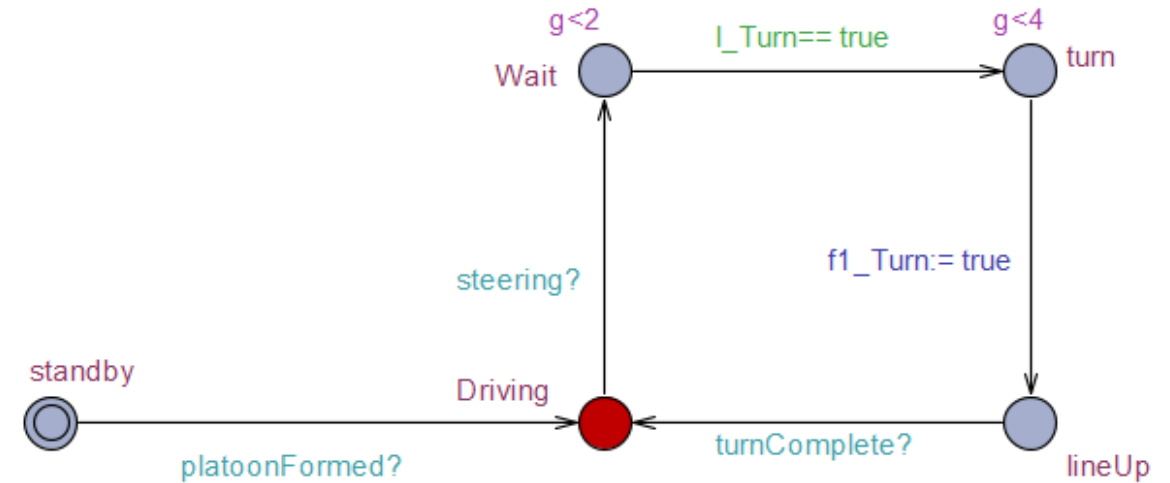


UPPAAL: FORMATION

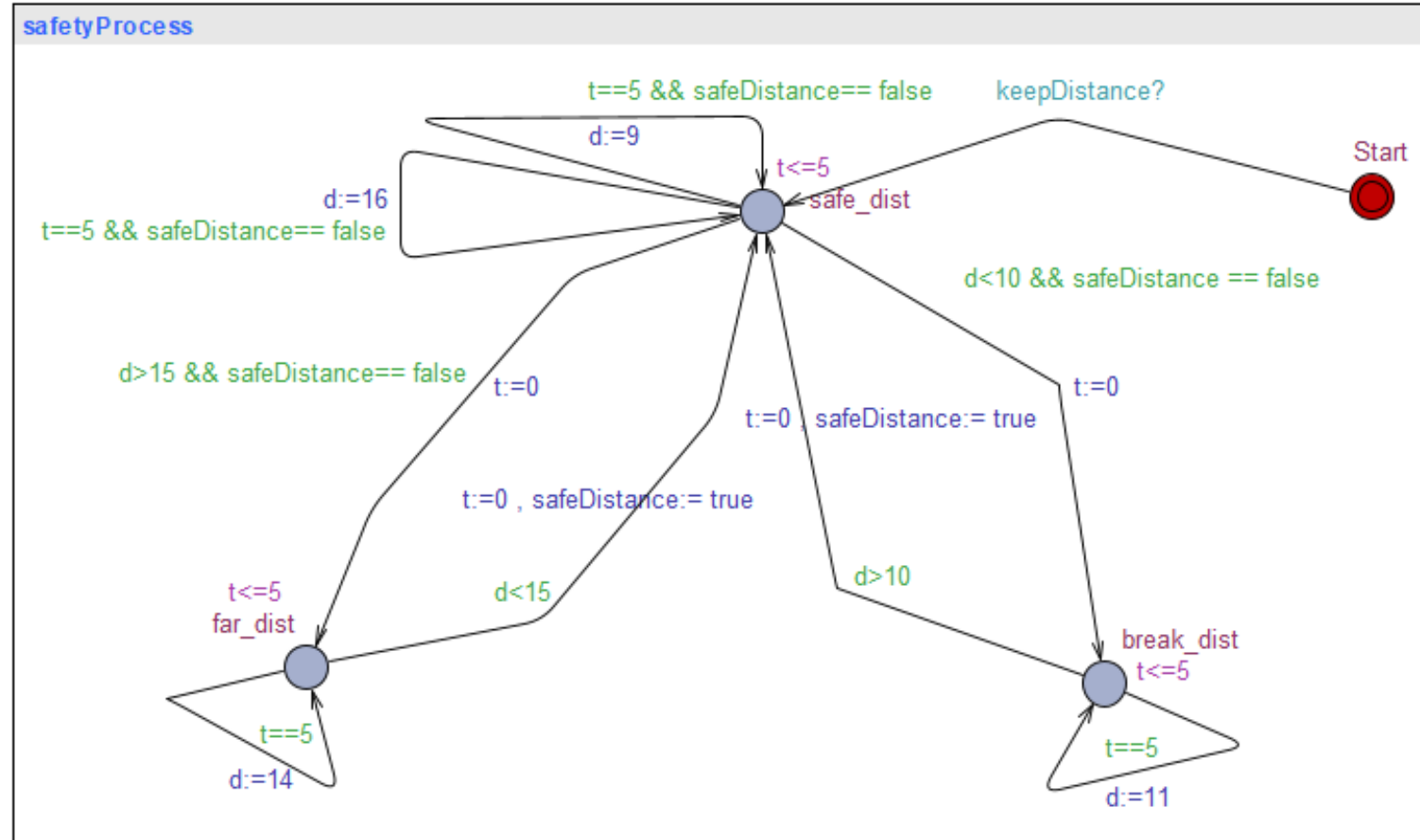
leader_ACC



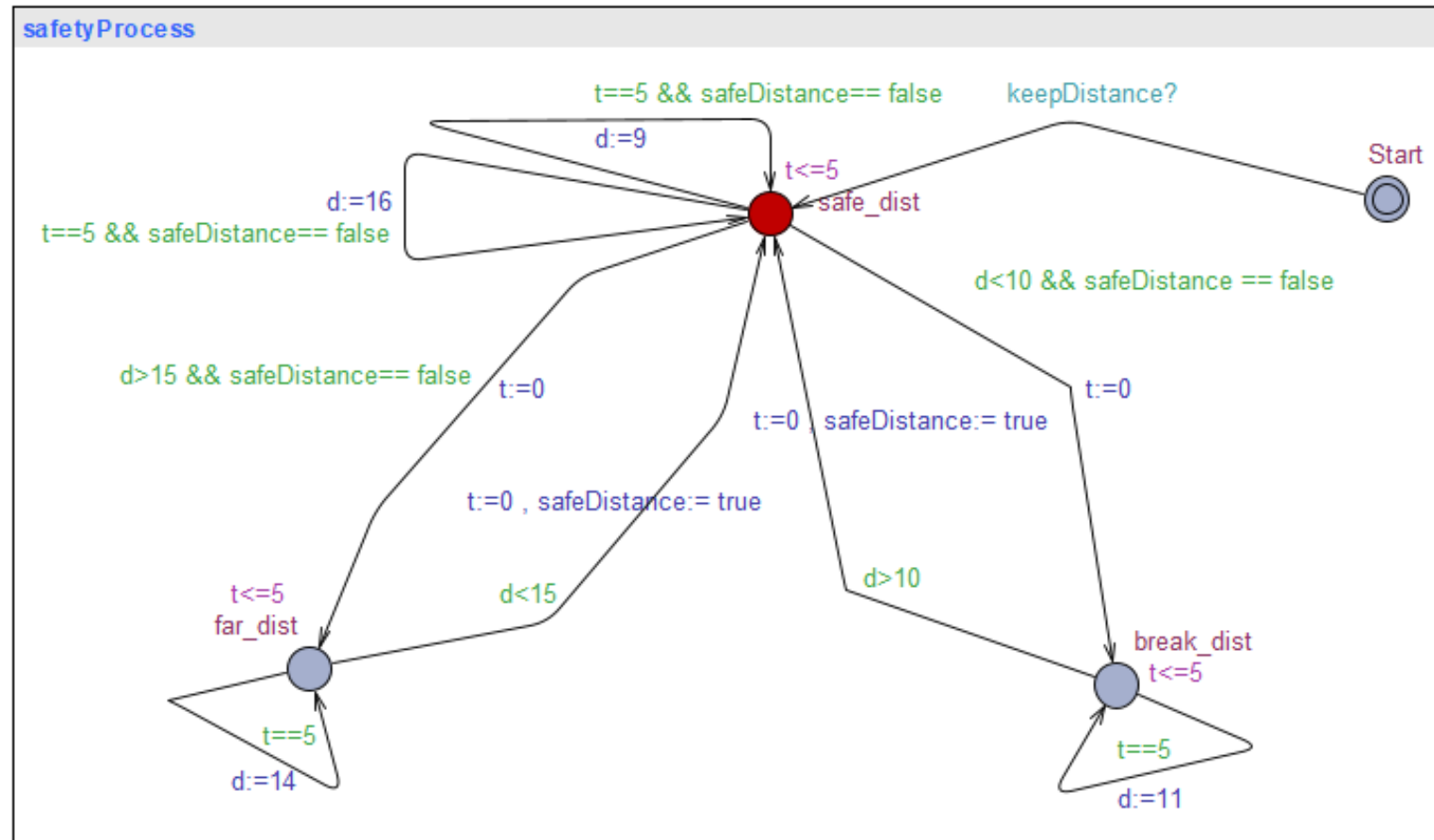
follower1_ACC



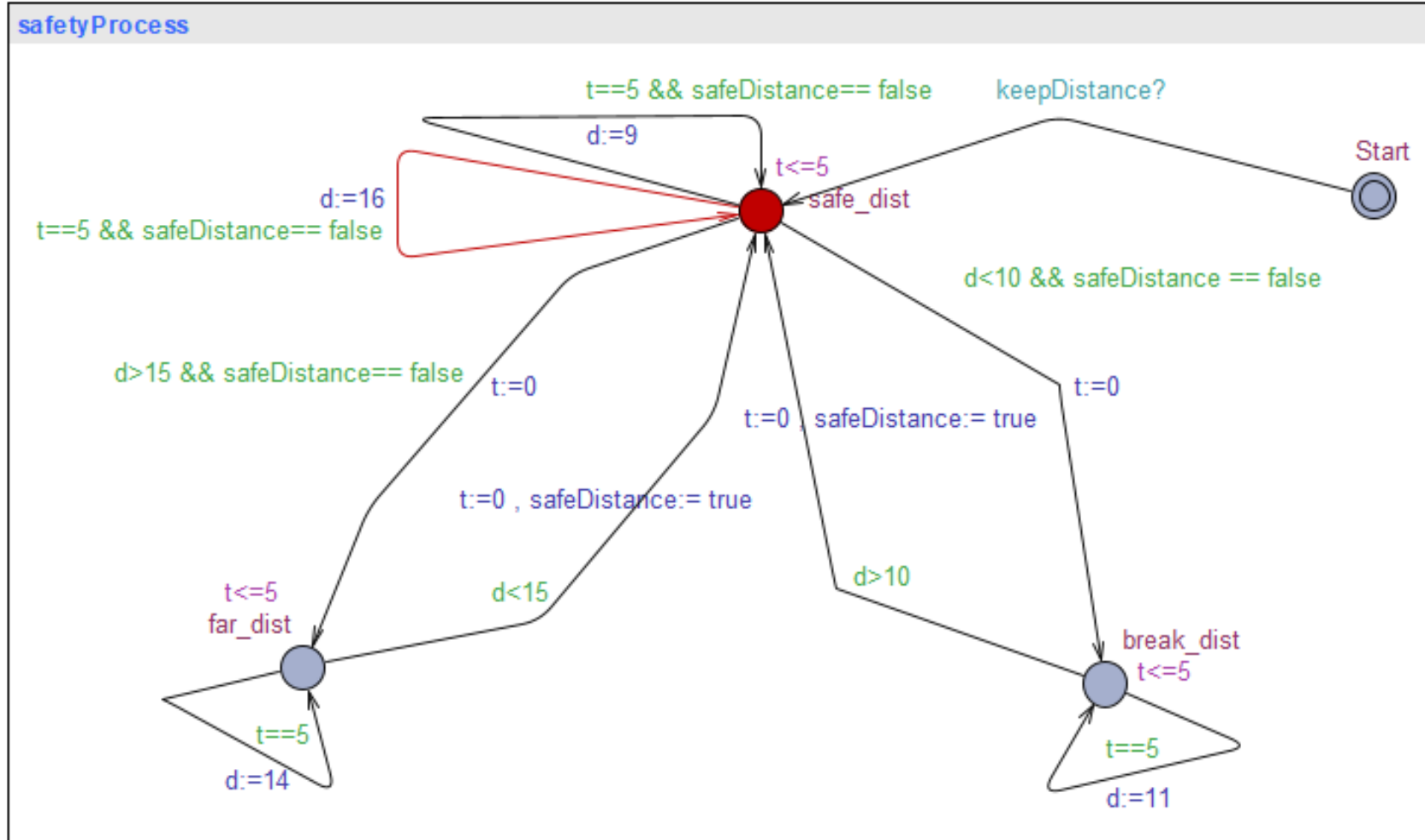
UPPAAL: SAFETY



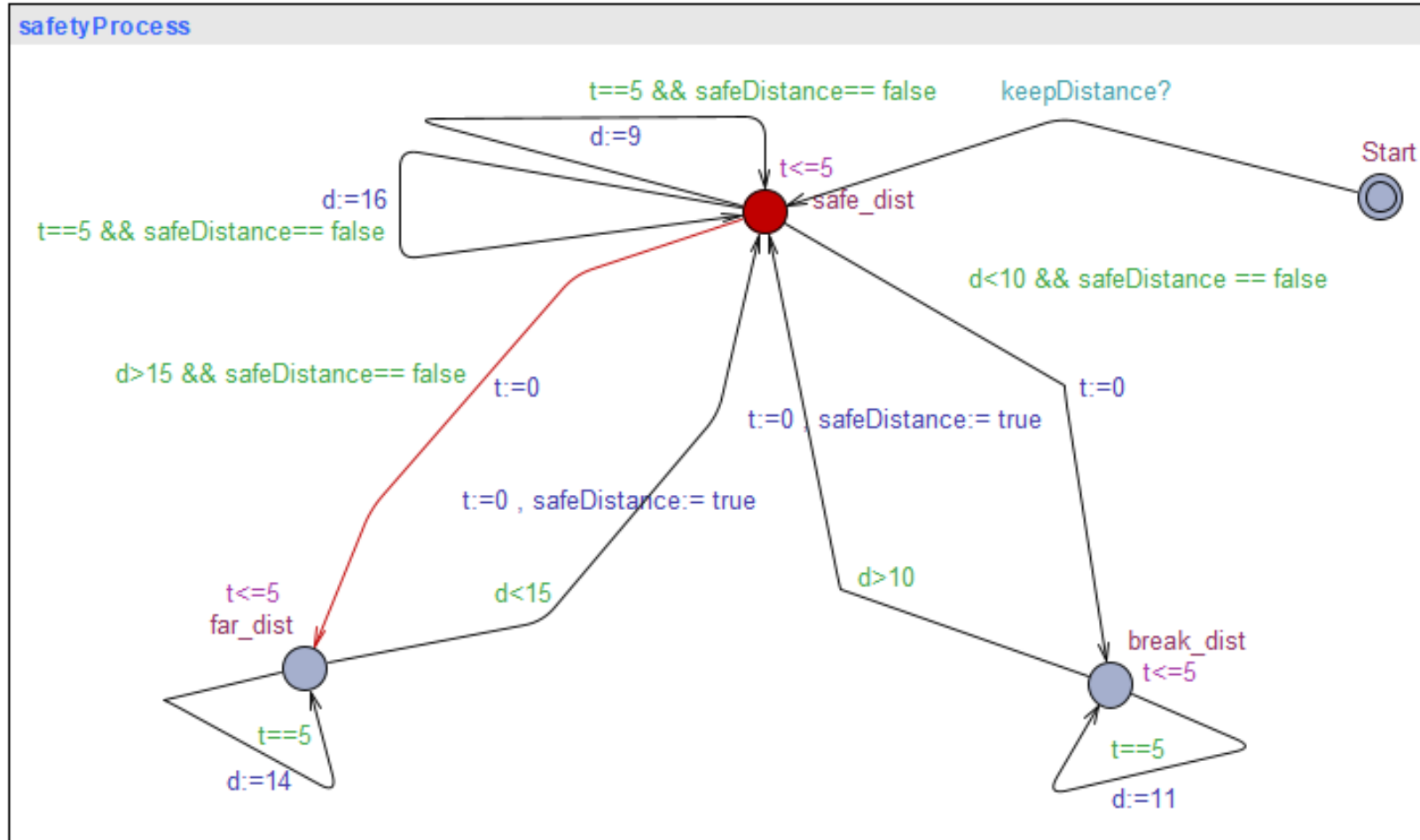
UPPAAL: SAFETY



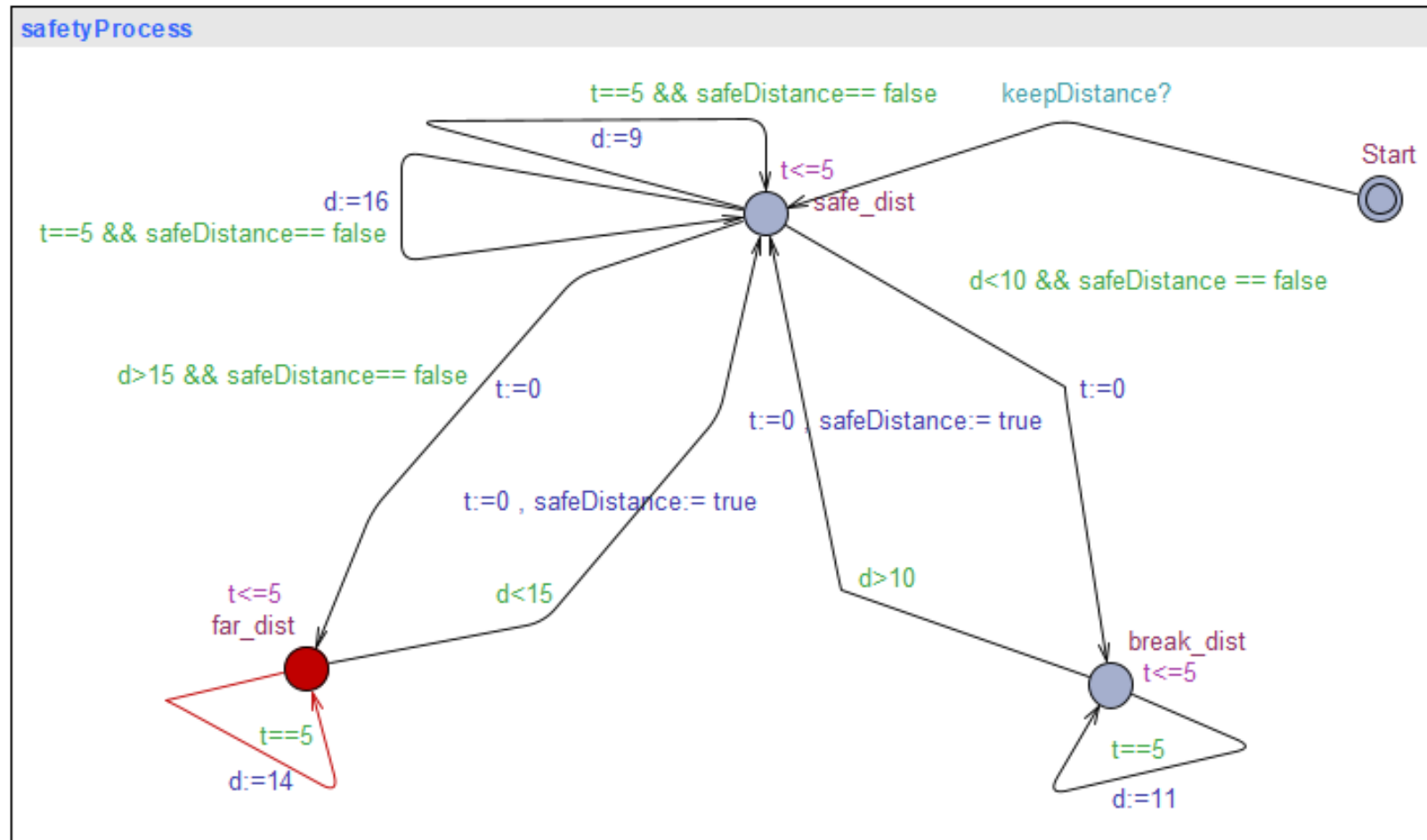
UPPAAL: SAFETY



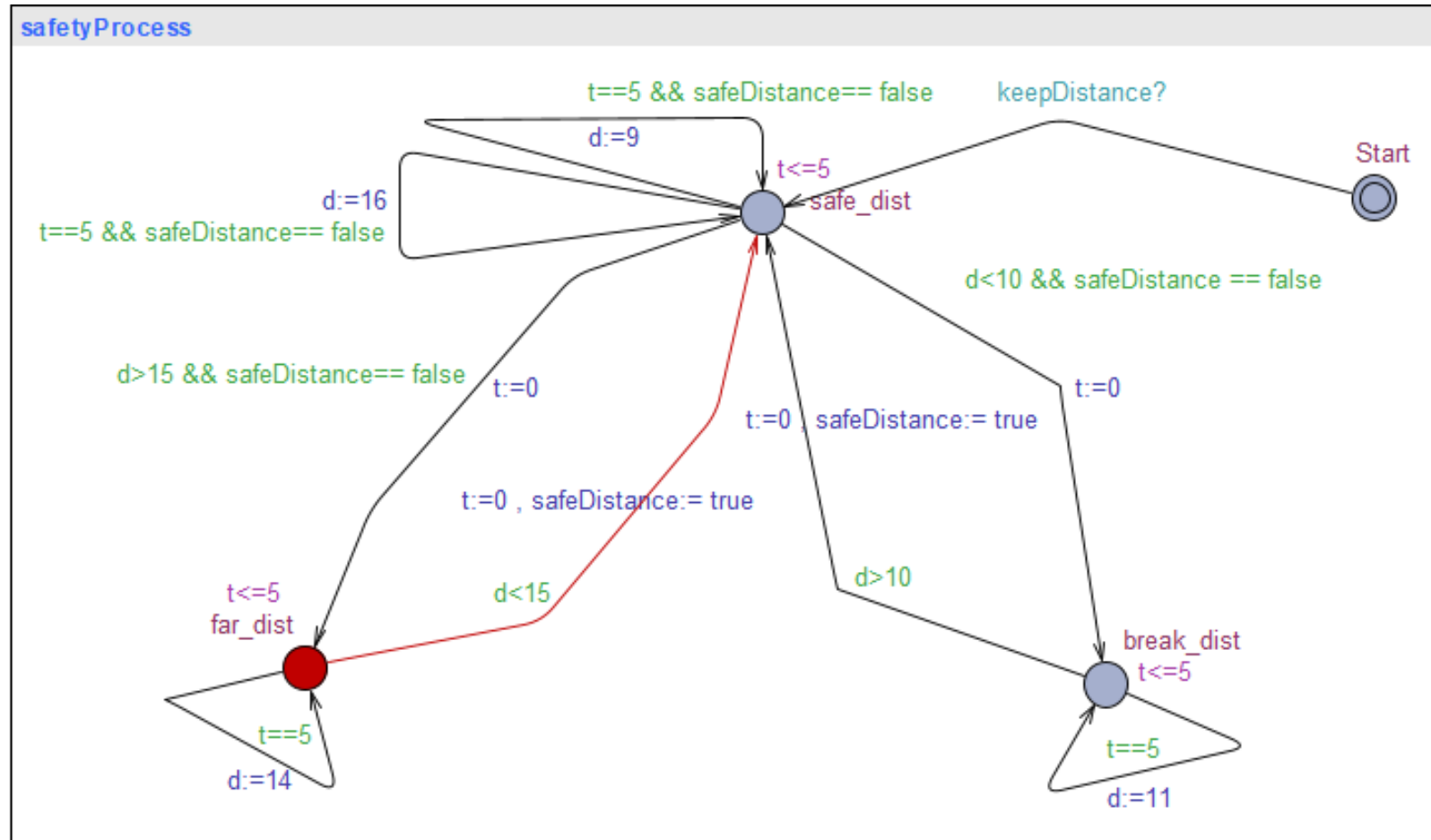
UPPAAL: SAFETY



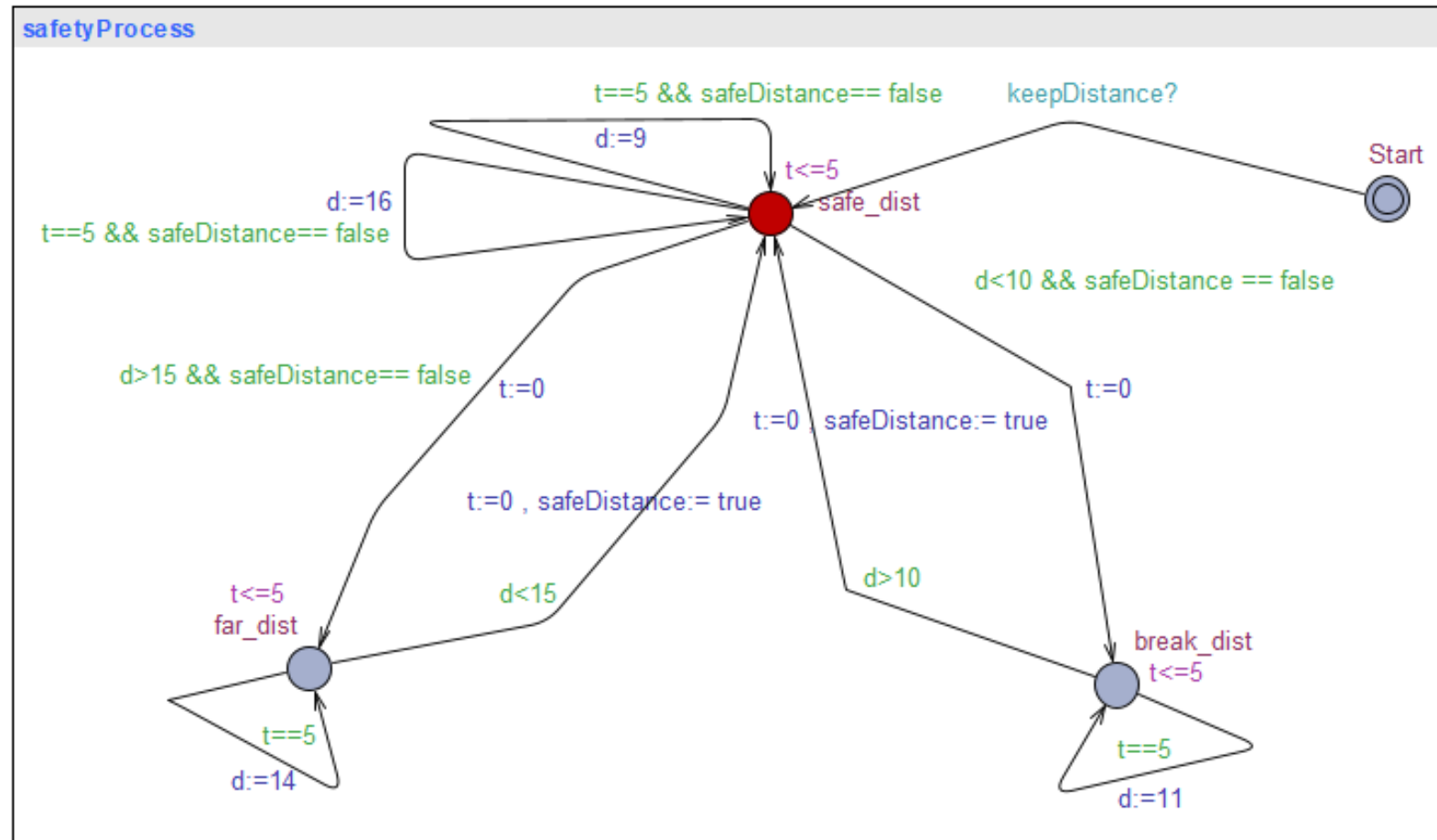
UPPAAL: SAFETY



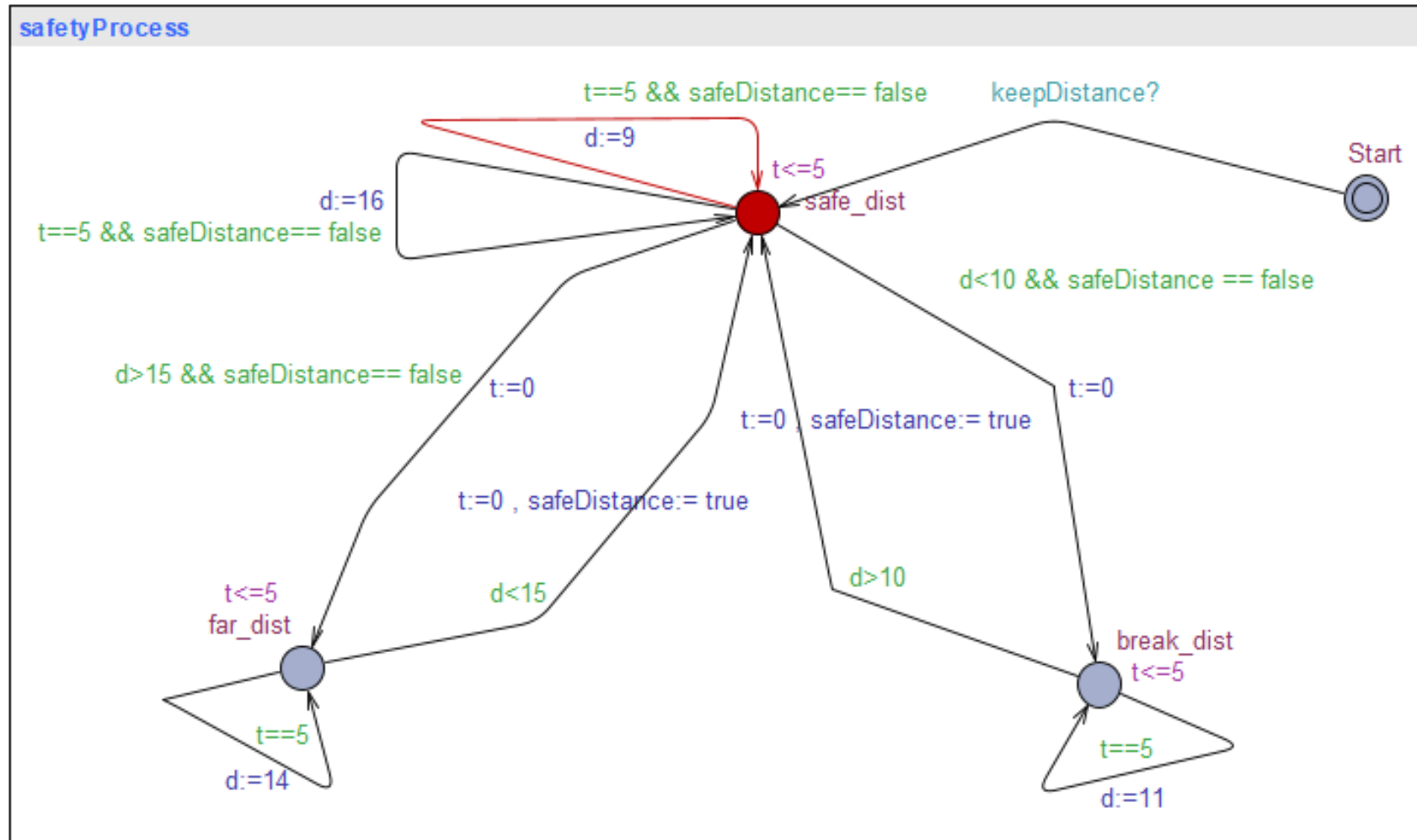
UPPAAL: SAFETY



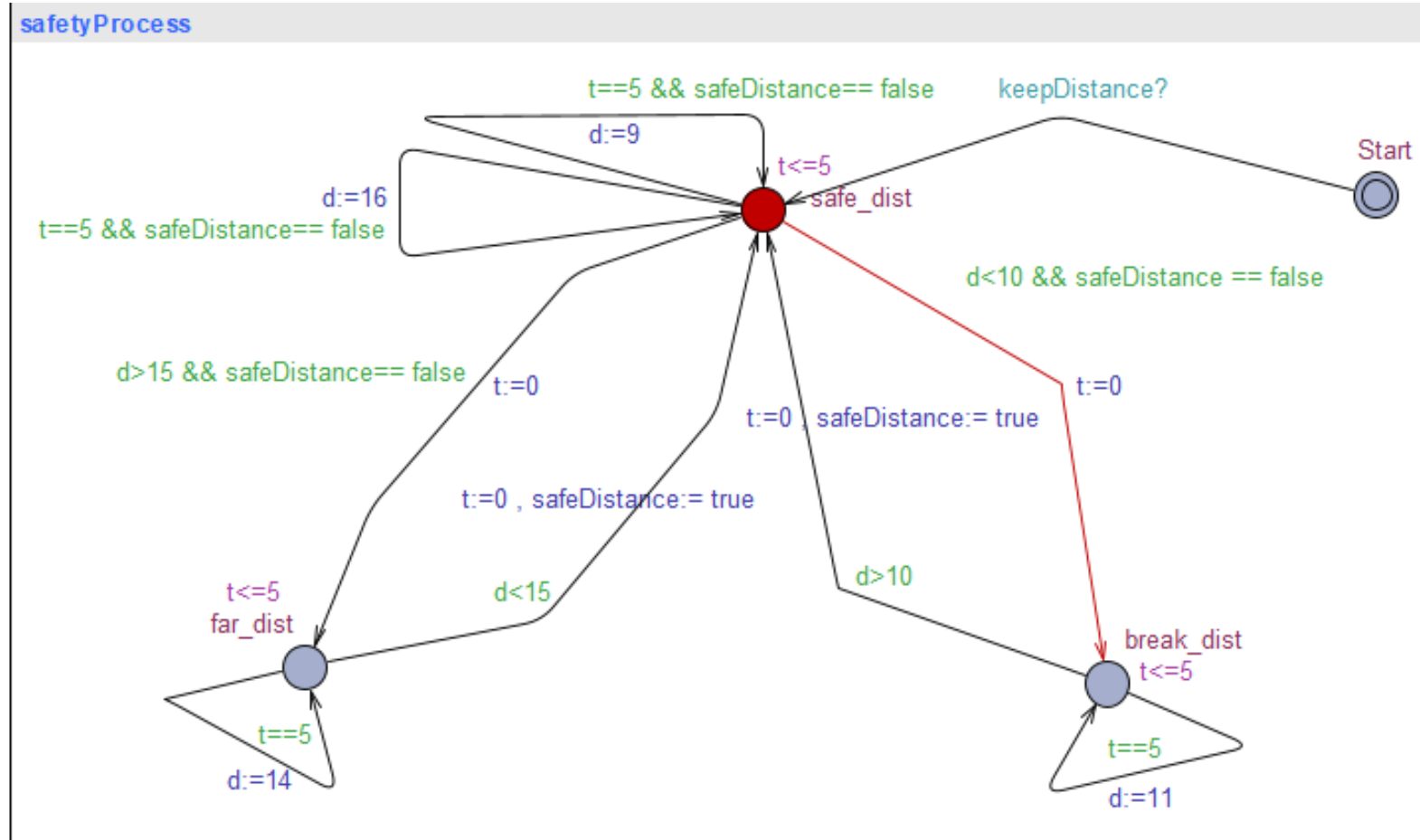
UPPAAL: SAFETY



UPPAAL: SAFETY

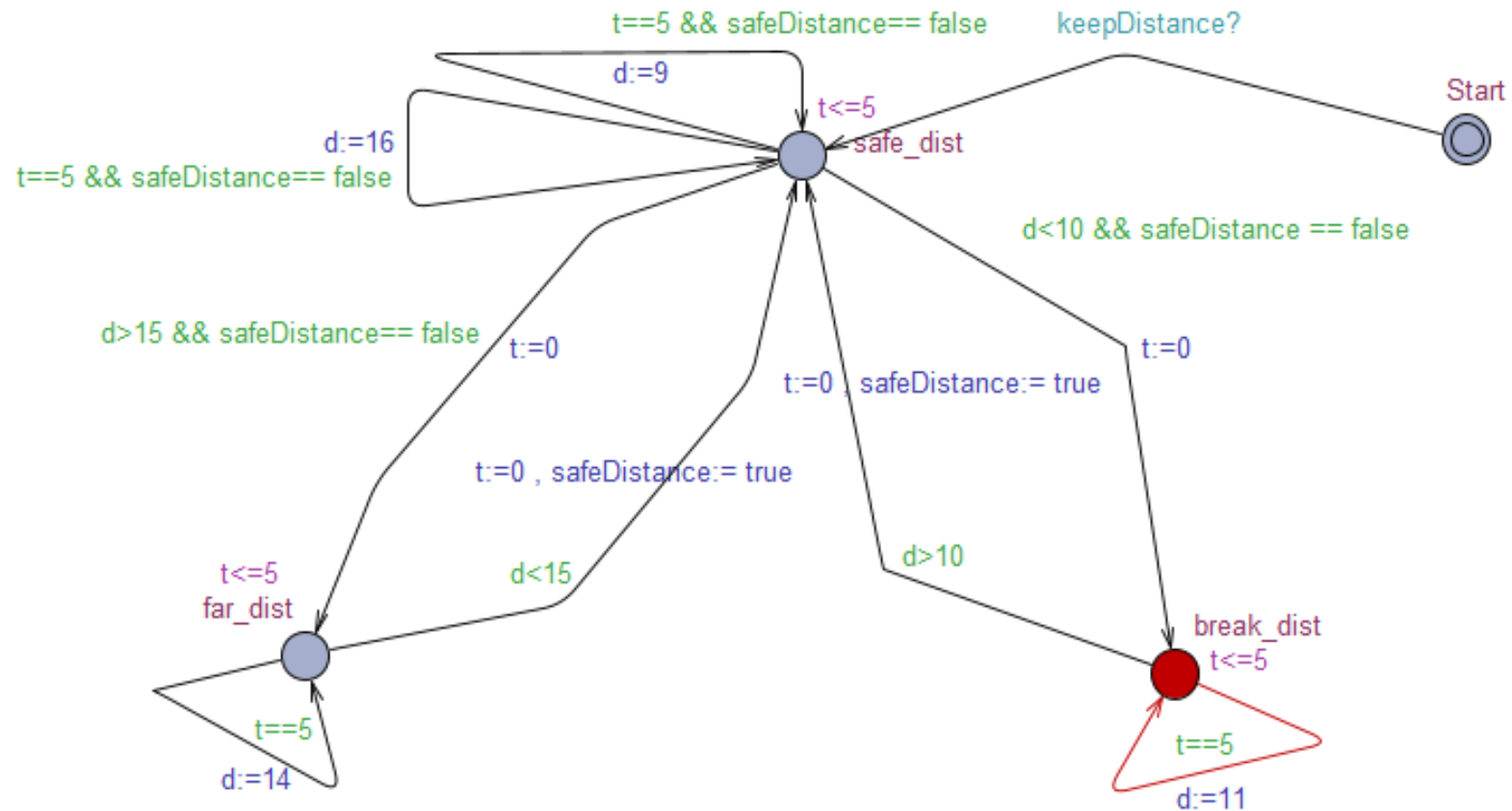


UPPAAL: SAFETY



UPPAAL: SAFETY

safetyProcess





UPPAAL: CONNECT

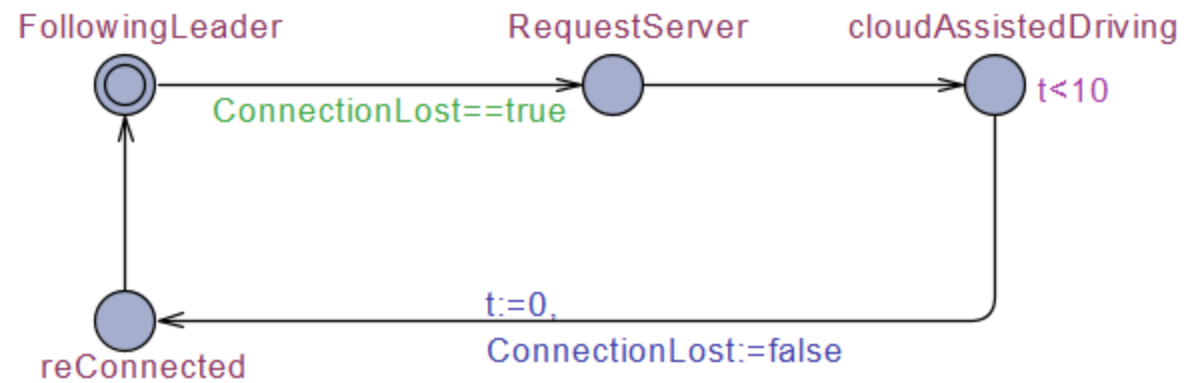


Fig: Uppaal Model

ML: DATA PREPARATION

years_of_experience	aerodynamics_score	technological_score
4	45	10
6	29	7
8	34	9
5	50	9
4	19	8
7	12	10
20	29	2
5	9	8
16	10	5
11	23	8
11	7	10
2	45	4
14	25	4
1	38	3
5	17	5

➤ Parameters

- years_of_experience = (0, 20)
- aerodynamics_score = (5, 50)
- technological_score = (1, 10)
- 5000 samples

ML: DATA PREPARATION

How do we label the data?

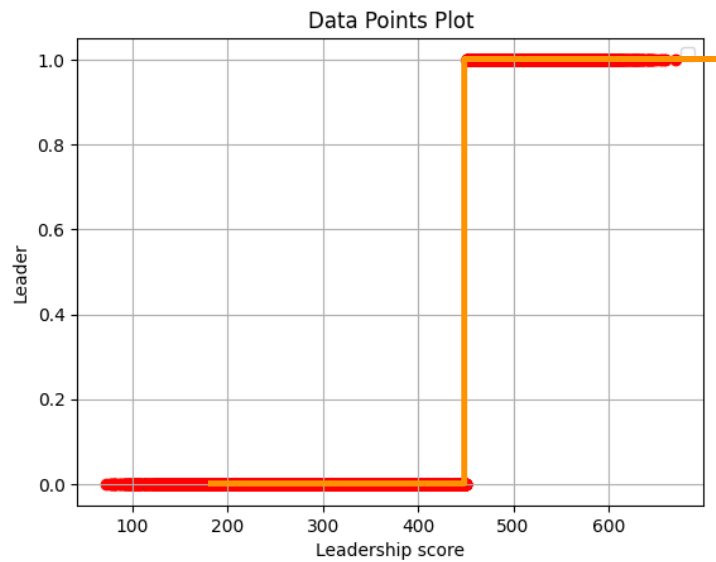
years_of_experience	aerodynamics_score	technological_score	Leadership Score	Leader
4	45	10	540	1
6	29	7	369	0
8	34	9	443	0
5	50	9	588	1
4	19	8	266	0
7	12	10	225	0
20	29	2	404	0
5	9	8	171	0
16	10	5	215	0
11	23	8	341	0
11	7	10	195	0
2	45	4	488	1
14	25	4	348	0
1	38	3	406	0
5	17	5	230	0

$$\text{Score} = w_1 \cdot A + w_2 \cdot B + w_3 \cdot C$$

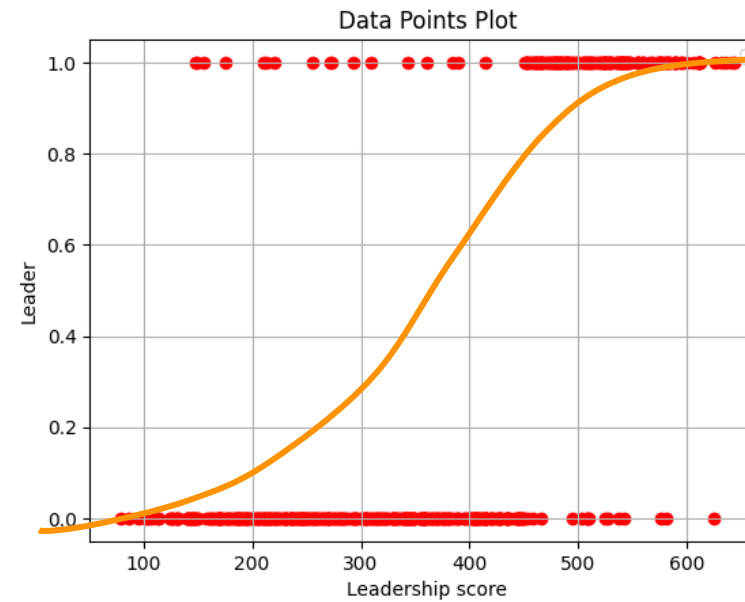
Where,

- $w_1 = 5; w_2 = 10; w_3 = 7;$
A = years_of_experience
B = aerodynamics_score
C = technological_score
- If score > 450 :
leader = True

ML: DATA PREPARATION



Over Fitting the Model



Added Outliers to avoid overfitting

ML:TRAINING

```
In [8]: from sklearn.model_selection import train_test_split  
X_train, X_test, Y_train, Y_test = train_test_split(X,Y,train_size=0.8)
```

```
In [9]: from sklearn.linear_model import LogisticRegression  
model = LogisticRegression()
```

```
In [10]: model.fit(X_train, Y_train)
```

ML: LEADER SELECTION

```
✓ [14] test = pd.read_csv("test.csv")  
0s test.head()
```

	years_of_experience	aerodynamics_score	technological_score
0	4	29	10
1	7	45	5
2	5	25	8

```
✓ [15] model.predict(test)  
0s  
array([0, 1, 0])
```

```
[22] test = pd.read_csv("test.csv")  
test.head()
```

	years_of_experience	aerodynamics_score	technological_score
0	4	29	10
1	7	45	5
2	15	35	8

```
[23] model.predict(test)  
array([0, 1, 1])
```

```
[24] probs = model.predict_proba(test)  
probs  
  
array([[0.78734264, 0.21265736],  
       [0.21864086, 0.78135914],  
       [0.40534463, 0.59465537]])
```

```
[25] probs_percent = probs * 100  
for row in probs_percent:  
    print(row[1])  
  
21.265736016383503  
78.13591381712006  
59.46553744303507
```

RANDOM FOREST CLASSIFIER

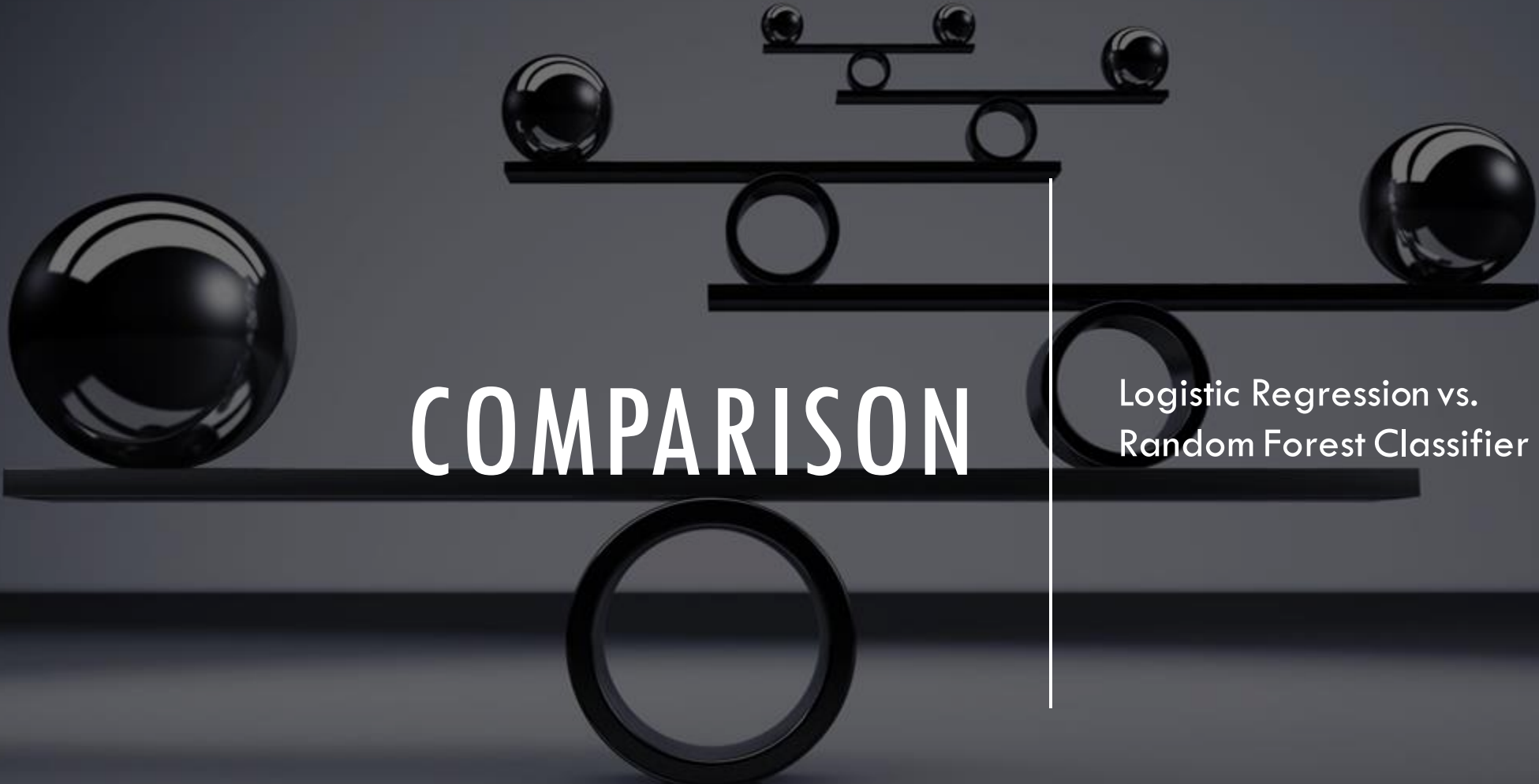
RANDOM FOREST CLASSIFIER

```
from sklearn.model_selection import train_test_split  
X_train, X_test, Y_train, Y_test = train_test_split(X,Y,train_size=0.8)
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
model = RandomForestClassifier()
```

```
model.fit(X_train, Y_train)
```



COMPARISON

Logistic Regression vs.
Random Forest Classifier

DECISION

Logistic regression

```
In [195]: model.score(X_test, Y_test)
```

```
Out[195]: 0.917
```

Random Forest Classifier

```
In [55]: model.score(X_test, Y_test)
```

```
Out[55]: 0.902
```

```
In [60]: for row in probs_percent:
          #index += 1
          if row[1] > leading_probability:
              leading_probability = row[1]
              Truck_num = index+1
          index += 1
```

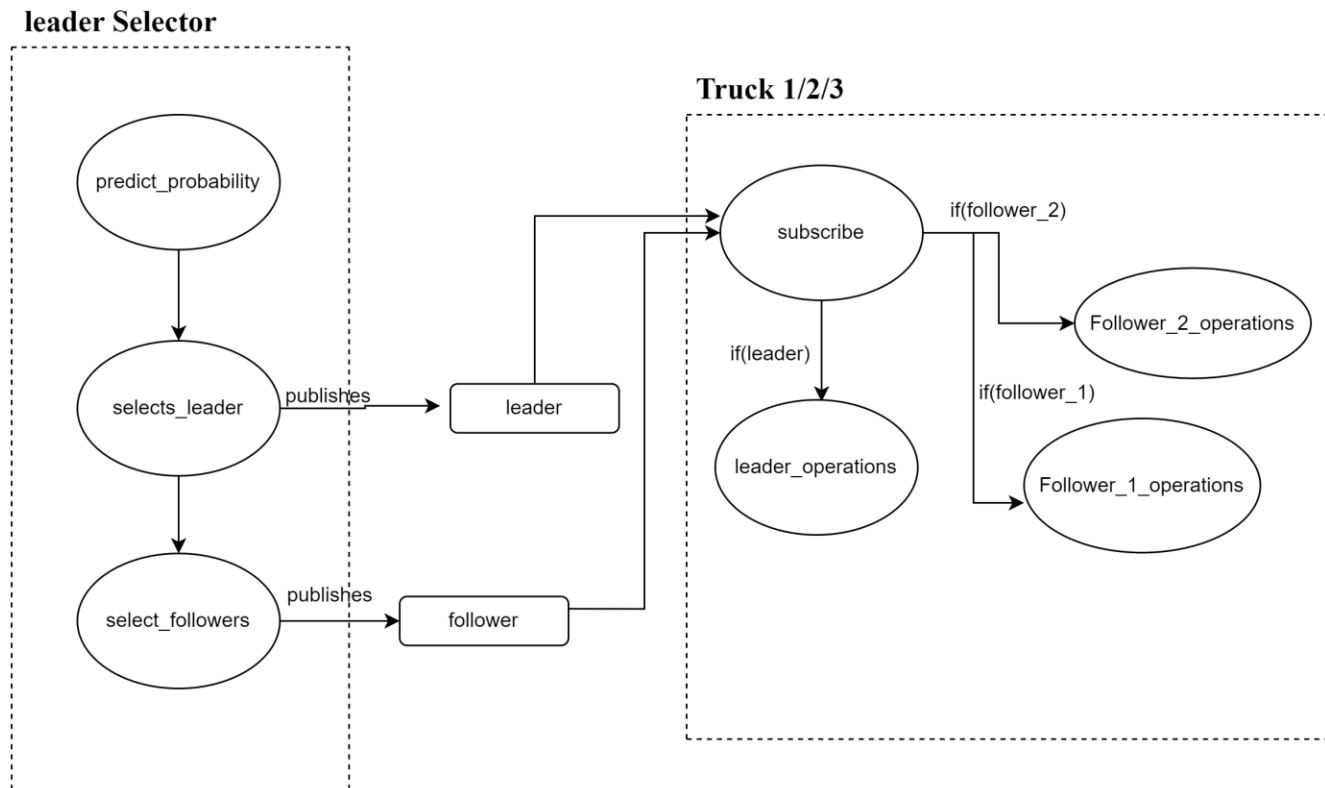
```
In [61]: Truck_num
```

```
Out[61]: 3
```

```
In [215]: Truck_num
```

```
Out[215]: 3
```

SIMULATION



SIMULATION

```
on) fahim@sahat:~/study/AutoSys/TruckNet-Gr-3/machine_learning_Model$ python3 leader_selector.py
```

Leader Selector

```
fahim@sahat:~/study/AutoSys/TruckNet-Gr-3/simulation$ python3 truck_
truck_1.py truck_2.py truck_3.py
fahim@sahat:~/study/AutoSys/TruckNet-Gr-3/simulation$ python3 truck_1.py
Connected to the Operator!!
Waiting for leader to be selected...
```

Truck 1

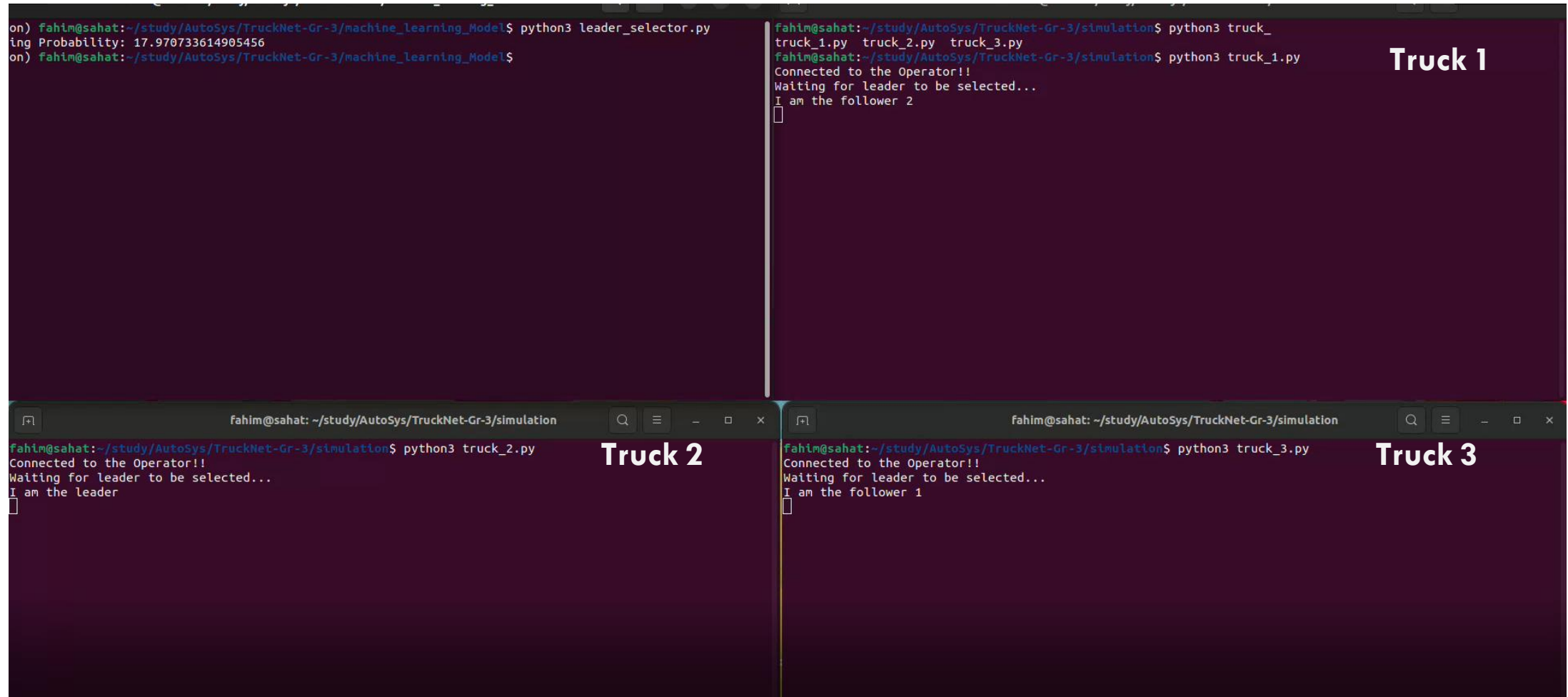
```
fahim@sahat:~/study/AutoSys/TruckNet-Gr-3/simulation$ python3 truck_2.py
Connected to the Operator!!
Waiting for leader to be selected...
```

Truck 2

```
fahim@sahat:~/study/AutoSys/TruckNet-Gr-3/simulation$ python3 truck_3.py
Connected to the Operator!!
Waiting for leader to be selected...
```

Truck 3

SIMULATION



```
on) fahim@sahat:~/study/AutoSys/TruckNet-Gr-3/machine_learning_Model$ python3 leader_selector.py
ing Probability: 17.970733614905456
on) fahim@sahat:~/study/AutoSys/TruckNet-Gr-3/machine_learning_Model$

fahim@sahat:~/study/AutoSys/TruckNet-Gr-3/simulation$ python3 truck_
truck_1.py truck_2.py truck_3.py
fahim@sahat:~/study/AutoSys/TruckNet-Gr-3/simulation$ python3 truck_1.py
Connected to the Operator!!
Waiting for leader to be selected...
I am the follower 2
[]

fahim@sahat:~/study/AutoSys/TruckNet-Gr-3/simulation$ python3 truck_2.py
Connected to the Operator!!
Waiting for leader to be selected...
I am the leader
[]

fahim@sahat:~/study/AutoSys/TruckNet-Gr-3/simulation$ python3 truck_3.py
Connected to the Operator!!
Waiting for leader to be selected...
I am the follower 1
[]
```

Truck 1

Truck 2

Truck 3



REFERENCES